

Medial-axis reconstruction on complete Riemannian manifolds: capture, drift, and the unconditional theorem*

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Abstract

For a closed set X in a complete Riemannian manifold M , the *two-geodesic medial axis* M_X^g is the set of points admitting two distinct minimising geodesics to X (the multi-geodesic part of the cut locus of X , in general a proper dense subset of it), and the reconstruction $\mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, d_X(a))$ is the union of its open medial balls. Across the series, the generalisation of Białożyty's Theorem 6 to Riemannian manifolds had been proved up to two flagged gaps: the total-reconstruction theorem on compact manifolds rested on an unproven extension of Lieutier's swallow property past cut points [5, Rem. 8.4], [6, Rem. 6.5], and the Hadamard characterisation for non-compact contact sets carried a radial-offset caveat [4, Thm. 3.4]. This note closes both, with no genericity, analyticity, curvature, or spread hypothesis. The first input is a *capture theorem*: along any solution of the differential inclusion $\Phi' \in -\text{conv } U(\Phi)$, where U is the (upper semicontinuous, in general non-closed-graph-selectable) field of unit directions of minimising geodesics to X , the distance d_X grows at unit rate for exactly as long as the trajectory avoids M_X^g ; hence every point of $M \setminus X$ is either *captured* — it lies at depth $\geq d_X(p)$ inside a medial ball of M_X^g — or its trajectory *escapes* to infinity, which is impossible when M is compact. Consequently $\mathcal{R}_X^g = M \setminus X$ for *every* compact M and every closed $X \neq \emptyset$ (even a single point). The subtlety that defeated the margin arguments — M_X^g is not closed, and the margin is not monotone through multi-geodesic points — is dissolved by stopping at the *first approach* to M_X^g rather than flowing through it; focal and conjugate degenerations are ordinary points of the inclusion. The second input is a horofunction formulation: calling ψ a *medial limit function* if $\psi = \lim [d(\cdot, a_k) - d_X(a_k)]$ locally uniformly with $a_k \in M_X^g$, the swallow lemma becomes three lines — $\{\psi < 0\} \subseteq \mathcal{R}_X^g$ — with no anchoring or pinching, and

$$\mathcal{R}_X^g = \bigcup \{ \{\psi < 0\} : \psi \text{ a medial limit function} \}$$

holds on *every* complete M . On a Hadamard manifold ideal medial limit functions are Busemann functions up to level, which resolves the offset caveat of [4]: a drift of offset c is exactly a medial limit at level c , reconstructing $\{b_\epsilon < -c\}$ — a statement, no longer a

*Final note of a series generalising Theorem 6 of [1]: [2] carried its sufficiency half to Hadamard manifolds, [3] proved the full characterisation in real hyperbolic space, [4] gave the faithful curvature-free characterisation for every Hadamard manifold, [5] crossed into positive curvature, and [6] organised the theory into its two axes (compactness; the Busemann–Hessian sign). This note removes the last analytic caveats and states the theorem on an arbitrary complete Riemannian manifold. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the two mechanics of the capture theorem — the rate-one identity off the axis and the margin at first approach — were verified numerically on $\mathbb{S}^2$ against a non-circular convex contact curve, whose focal region forces the foot to slide (60 random starts: growth rate 1 to five digits, margin deficit $< 3 \cdot 10^{-6}$; file `medial_capture_check.py`). Remaining errors are ours.

gap. Together with the trichotomy captured/drift-swallowed/escaping, this completes the programme: the Euclidean, Hadamard, spherical, spread, and compact theorems are all specialisations, and the remaining open problems are exactly the faithful (M_X^g -free) certificates on the escape side. As a first instance beyond the series we prove the end certificate on the flat cylinder: the free end-region above $\sup_X z$ is reconstructed in full if the top contact set omits a circle position, and not at all if it is the full circle. A closing section adds the homotopy layer: the escape class of the trichotomy is exactly the Aubry set of d_X in the sense of Cannarsa–Cheng–Fathi, so the captured part of the complement always has the homotopy type of M_X^g ; whenever $\sup d_X < \infty$ — every compact M , every coarsely dense X — capture supplies the uniform clock that boundedness of the domain supplied for Lieutier and Albano–Cannarsa–Nguyen–Sinestrari, and $M \setminus X \simeq M_X^g \simeq \text{Cut}(X)$. The cylinder with $|X| = 1$ shows the bound is not cosmetic: the reconstruction is complete, $\mathcal{R}_X^g = C \setminus X$, yet a figure-eight complement faces a contractible axis — homotopy equivalence follows capture, and swallowing does not transmit it. A final section repairs the leak, unconditionally: every swallowed-uncaptured point lies in an *ideal* free horoball, at depth no smaller than its depth in any finite medial ball — the finite layer does not leak — so $\mathcal{R}_X^g$ is the captured set plus the ideal cells, and the *medial complex*, M_X^g with the ideal free horoballs glued along the diagram of the cover, always carries the homotopy type of $\mathcal{R}_X^g$, of $M \setminus X$ when the reconstruction is complete: on the cylinder it rebuilds the figure-eight. The complex is not a curvature tax: three points in the plane already reconstruct completely while their complement, a wedge of three circles, faces a contractible Voronoi tripod — the flat no-leak conjecture is false, the leak sitting in the corner cones where two ideal half-planes overlap inside the Aubry set. Two classical instances and a smooth refinement close the layer: uniform capture recovers the Ford–Voronoi spines of finite-volume cusped hyperbolic manifolds and the Voronoi spines of crystallographic quotients; and for a smooth closed submanifold X the captured part of the complement is, always, the mapping cylinder of the swallow map $S(\nu) \rightarrow \text{Cu}(X)$ — the homotopy equivalence sharpens to a strong deformation retraction onto the full cut locus, with the Thom-space collapse of Basu–Prasad as its inward dual.

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1 Introduction

For a closed set $X \neq \emptyset$ in a complete Riemannian manifold M write $d_X = \text{dist}(\cdot, X)$,

$$M_X = \{a : a \text{ has } \geq 2 \text{ nearest points in } X\}, \quad M_X^g = \{a : a \text{ has } \geq 2 \text{ minimising geodesics to } X\},$$

for the two-*point* and two-*geodesic* medial axes ($M_X \subseteq M_X^g$; the endpoints of the two geodesics may coincide), and

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, d_X(a)), \quad \mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, d_X(a))$$

for the corresponding reconstructions by open medial balls. Białożyty's Theorem 6 [1] characterises $\mathcal{R}_X$ in $\mathbb{R}^n$ (where $M_X = M_X^g$); the companion notes carried the characterisation to real hyperbolic space [3], to every Hadamard manifold with a faithful horoconvexity certificate [2, 4], and into positive curvature [5], and [6] organised the results into two independent axes: *compactness* governs partial-versus-total, the *Busemann–Hessian sign* governs the detection flavour.

The two inputs

Capture (§3). The obstruction in the first flagged gap — the containment step past cut points on compact manifolds [5, Rem. 8.4], [6, Rem. 6.5] — was never the existence of the ascending flow (on Riemannian domains its canonical, semiconcave form is the singular gradient flow of Albano–Cannarsa–Nguyen–Sinestrari [9]; see Remark 3.5); it was the bookkeeping of the margin through multi-geodesic points, where the flow turns and the margin genuinely need not be monotone (Remark 3.4). The repair is to reverse the roles: follow *any* trajectory of the differential inclusion driven by the minimising-geodesic directions, and *stop at the first approach to M_X^g* . Off M_X^g the direction field is single-valued, the trajectory is a unit-speed curve along which d_X grows at rate exactly one — through focal and conjugate points without comment — so

$$d_X(\Phi_t) = d_X(p) + t, \quad d(p, \Phi_t) \leq t,$$

and the starting point sits at depth $\geq d_X(p)$ in every moving ball. The trajectory therefore either approaches M_X^g in finite time — *capture*: some $a \in M_X^g$ has $d(p, a) < d_X(a)$, i.e. $p \in \mathcal{R}_X^g$ — or avoids M_X^g forever and $d_X \rightarrow \infty$ along it — *escape*, impossible on a compact manifold. The fact that M_X^g is not closed (minimising geodesics can merge in the limit: focal degenerations) is harmless, because approach suffices: the margin is continuous. This is Theorem 3.1, and it makes the compact theorem unconditional (Corollary 3.2), for every closed $X \neq \emptyset$ — including $|X| = 1$, which the two-point axis cannot see at all. In particular the reconstruction question on compact Riemannian manifolds, raised for the medial axis in [1, §5.2], has an unconditional affirmative answer for the two-geodesic axis.

Drift (§4). Capture cannot be the whole story on a non-compact manifold: a point can be reconstructed by medial balls whose centres run to infinity while its own trajectory escapes (the open half-planes of two points in $\mathbb{R}^2$ are the primal example). The right limit object costs nothing in a complete manifold: M is proper (Hopf–Rinow), so the functions $\psi_a = d(\cdot, a) - d_X(a)$ — the *free-ball functions* of medial centres, normalised by their own reach, with no basepoint — are precompact in the topology of locally uniform convergence along any sequence that reconstructs a fixed point. Calling any such limit a *medial limit function*, the swallow lemma of [4] becomes three curvature-free lines with no anchoring hypothesis: $\{\psi < 0\} \subseteq \mathcal{R}_X^g$ for every medial

limit function ψ (Lemma 4.1). The characterisation on an arbitrary complete manifold follows (Theorem 4.3), together with the trichotomy: every $p \in M \setminus X$ is captured, drift-swallowed, or escaping-unswallowed, and $\mathcal{R}_X^g$ is exactly the first two classes (Theorem 4.4). On a Hadamard manifold the ideal medial limit functions are Busemann functions up to an additive level, and the radial-offset caveat of [4, Thm. 3.4] dissolves into a definition: a drift with asymptotic offset c is a medial limit function at level c and reconstructs $\{b_\xi < -c\}$ in full (Remark 4.6).

What the completed theory says

Section 5 re-derives every earlier result as a specialisation and updates the 2×2 of [6]: the compact row is unconditionally total for $\mathcal{R}_X^g$; the non-compact row is governed by the medial limit functions, whose faithful (M_X^g -free) certificates are known exactly on the geometries where the series established them — conv in $\mathbb{R}^n$ [1], hconv on Hadamard manifolds [4], vacuous on compact M — and open in mixed curvature, with the flat cylinder settled here as the first instance beyond them (Proposition 5.1). Those certificates, and the quantitative cut-shielding threshold for the two-point axis [6, Rem. 6.6], are the honest residue of the programme (§8).

Section 6 adds the topological counterpart of the reconstruction theory. The escape class of the trichotomy turns out to be exactly the *Aubry set* of d_X in the sense of Cannarsa–Cheng–Fathi [15], so their theorem reads: the captured part of the complement always has the homotopy type of M_X^g . When $\sup d_X < \infty$ — every compact M , every coarsely dense X — capture supplies the uniform clock that boundedness of the domain supplied in [25, 9], and $M \setminus X \simeq M_X^g \simeq \text{Cut}(X)$ on every complete manifold, components of $M \setminus X$ unbounded or not. The flat cylinder separates the metric and topological completions: $\mathcal{R}_X^g = C \setminus X$ in full, yet $C \setminus X \not\simeq M_X^g$ — homotopy follows capture, not swallowing. The section closes with the classical geometry this contains: uniform capture recovers the Ford–Voronoi spines of finite-volume cusped hyperbolic manifolds — dual to the Epstein–Penner decomposition [20] — and the Voronoi spines of crystallographic quotients (Corollary 6.7); and for a smooth closed submanifold the captured part of the complement is, with no further hypothesis, the mapping cylinder of the swallow map onto the cut locus — a strong deformation retraction, with the Basu–Prasad Thom-space picture as its inward dual (Proposition 6.8, Remark 6.9).

Section 7 then closes the leak. Its engine is a covering theorem with no counterpart in the series: *the finite layer does not leak* — every swallowed-uncaptured point lies in an *ideal* free horoball, at depth no smaller than its depth in any finite medial ball (Theorem 7.3). The proof runs the capture clock along the transition between the captured set and the Aubry set: either the clocks of nearby captured points blow up, dragging medial centres to infinity with the margin intact, or the transition ray grazes the cut locus — and then, by the invariance of Lemma 6.1, rides it to infinity, with the same effect. Consequently $\mathcal{R}_X^g$ is the captured set plus the ideal cells (Corollary 7.4), and the *medial complex* — M_X^g with the ideal free horoballs glued along the diagram of the cover — always carries the homotopy type of $\mathcal{R}_X^g$, of $M \setminus X$ when the reconstruction is complete (Theorem 7.5): on the cylinder it rebuilds the figure-eight from a contractible axis and two ideal circles. The escaping class becomes ideally visible too — class (3) is exactly the Aubry points on which every ideal medial limit function is nonnegative — and when X has no ideal medial limit functions at all, $\mathcal{R}_X^g \simeq M_X^g$ with no bound on d_X (Corollary 7.6). The section ends by settling the flat no-leak conjecture of the earlier draft, negatively: three points in the plane reconstruct completely and leak all of their homotopy, the loops living in the corner cones where two ideal half-planes overlap inside the Aubry set (Proposition 7.8); for finite planar contact sets the leak has rank exactly the number of convex-hull corners (Remark 7.9).

2 Preliminaries

Throughout, M is a complete Riemannian manifold (hence a proper geodesic metric space, by Hopf–Rinow), $X \subseteq M$ is closed with $\emptyset \neq X \neq M$, and $d_X = \text{dist}(\cdot, X)$. All geodesics are unit-speed. We state the standing hypotheses once, to prevent a misreading the series invites: *complete metric, closed X — nothing else*. No curvature assumption of any kind (sign, constancy, pinching, bounds) appears in §2–§4; the curvature of M may vary and change sign freely. For $x \in M \setminus X$ let

$$U(x) = \{\gamma'(0) : \gamma : [0, d_X(x)] \rightarrow M \text{ minimising geodesic, } \gamma(0) = x, \gamma(d_X(x)) \in X\} \subseteq T_x^1 M,$$

the set of unit initial directions of minimising geodesics to X . Since a geodesic is determined by its initial vector, distinct minimising geodesics have distinct initial directions, so

$$M_X^g = \{x \in M \setminus X : |U(x)| \geq 2\}.$$

We write $\Sigma := M_X^g$ when we use it as a stopping set, and $r(a) = d_X(a)$ for the reach of a centre.

Lemma 2.1 (Direction field: compactness and semicontinuity). *For every $x \in M \setminus X$ the set $U(x)$ is nonempty and compact, and the multimap $x \mapsto U(x)$ is upper semicontinuous: if $x_k \rightarrow x$ in $M \setminus X$ and $u_k \in U(x_k)$, $u_k \rightarrow u$ (in TM), then $u \in U(x)$. In particular U is single-valued and continuous at every point of $(M \setminus X) \setminus \Sigma$.*

Proof. Nonemptiness and compactness: minimisers exist by properness, and a limit of initial vectors of minimising geodesics from the same x is the initial vector of a limit geodesic, which minimises by continuity of d_X and of lengths. Upper semicontinuity: let γ_k be the minimising geodesics with $\gamma_k'(0) = u_k$. By smooth dependence of geodesics on initial conditions, $\gamma_k \rightarrow \gamma$ locally uniformly where $\gamma'(0) = u$; the lengths $d_X(x_k) \rightarrow d_X(x)$, and the endpoints $\gamma_k(d_X(x_k)) \in X$ converge to $\gamma(d_X(x)) \in X$ (X closed). So γ is a geodesic from x to X of length $d_X(x)$: minimising, i.e. $u \in U(x)$. At a point with $U(x) = \{u\}$, upper semicontinuity of a compact-valued multimap into a singleton forces continuity of every selection near x — note this is continuity at x only; U may be multivalued arbitrarily close by. $\square$

Lemma 2.2 (First variation of d_X ; [26, §3]). *For $x \in M \setminus X$ and $w \in T_x M$, the one-sided directional derivative of d_X exists and equals*

$$\left. \frac{d}{ds} \right|_{s=0^+} d_X(\exp_x(sw)) = \min_{u \in U(x)} \langle w, -u \rangle.$$

Proof. “ $\leq$ ”: fix $u \in U(x)$ with geodesic γ , and fix a small $\delta > 0$. The point $\gamma(\delta)$ lies in a normal ball around x , so $q \mapsto d(q, \gamma(\delta))$ is smooth near x with gradient $-u$ at x . Since points of γ satisfy $d_X(\gamma(\delta)) = d_X(x) - \delta$,

$$d_X(\exp_x(sw)) \leq d(\exp_x(sw), \gamma(\delta)) + d_X(x) - \delta = d_X(x) + s \langle w, -u \rangle + O_\delta(s^2),$$

so $\limsup_{s \downarrow 0} s^{-1}[d_X(\exp_x(sw)) - d_X(x)] \leq \langle w, -u \rangle$ for every $u \in U(x)$, hence $\leq$ the minimum.

“ $\geq$ ”: let $s_k \downarrow 0$ realise the $\liminf$, put $x_k = \exp_x(s_k w)$, pick $u_k \in U(x_k)$ with geodesics γ_k , and set $\delta_k = \sqrt{s_k}$. The broken path through $\gamma_k(\delta_k)$ gives $d_X(x) \leq d(x, \gamma_k(\delta_k)) + d_X(x_k) - \delta_k$, and in normal coordinates at x_k , where $v_k := \exp_{x_k}^{-1}(x) = -s_k w + O(s_k^2)$,

$$d(x, \gamma_k(\delta_k)) \leq |\delta_k u_k - v_k| + O(\delta_k^3) \leq \delta_k - \langle v_k, u_k \rangle + \frac{|v_k|^2}{2\delta_k} + O(\delta_k^3) = \delta_k + s_k \langle w, u_k \rangle + O(s_k^{3/2}),$$

using $\sqrt{1+\tau} \leq 1 + \tau/2$ and the $O(r^2)$ metric distortion of normal coordinates. Hence $d_X(x_k) - d_X(x) \geq s_k \langle w, -u_k \rangle - O(s_k^{3/2})$. By Lemma 2.1 a subsequence of u_k converges to some $u \in U(x)$, so the $\liminf$ is $\geq \langle w, -u \rangle \geq \min_{u \in U(x)} \langle w, -u \rangle$. (Cf. [26, §3], where the semiconcavity of d_X on $M \setminus X$ — not needed here — is also established.) $\square$

Prior art on the distance from a closed set. The dichotomy beneath everything here — d_X is differentiable exactly where the foot direction is unique, singular exactly on Σ — is classical, and its fine structure theory is active. Tanaka and Sabau [31] characterise the differentiable points of the distance from an arbitrary closed subset in Finsler generality and prove that on surfaces its cut locus is a local tree, statements new even in the Riemannian case; Miura and Tanaka [27] prove a delta-convex structure theorem for the singular set. The capture argument needs none of this finer structure — only Lemmas 2.1 and 2.2 below — but the structure theory is what a regularity upgrade of the homotopy layer of §6, and a CW model of the medial complex of §7 (Remark 7.7(c)), would draw on.

The descent inclusion. On the open set $M \setminus X$ define the multimap

$$G(x) := \text{conv}(-U(x)) \subseteq T_x M.$$

By Lemma 2.1, G is upper semicontinuous with nonempty compact convex values contained in the closed unit ball. A *capture curve* from $p \in M \setminus X$ is a locally Lipschitz curve $\Phi : [0, T_{\max}) \rightarrow M \setminus X$ with $\Phi_0 = p$ and

$$\Phi'(t) \in G(\Phi_t) \quad \text{for a.e. } t, \tag{1}$$

maximally extended in $M \setminus X$. Local existence of solutions of (1), and extension as long as the trajectory stays in a compact subset of the open domain, are the standard theory of differential inclusions with upper semicontinuous convex compact right-hand side (Aubin–Cellina [7, §2.1]), applied in charts and concatenated; $|\Phi'| \leq 1$ a.e., so trajectories are 1-Lipschitz. We emphasise that *no* uniqueness, canonicity, or semiconcave-gradient selection is needed: any solution serves. (At a point of Σ the inclusion may branch; off Σ it is the classical ODE of the field $-u$.) When U is single-valued along a trajectory the curve simply extends the current minimising geodesic beyond x , switching fields continuously as the foot slides.

Free-ball functions. For $a \in M \setminus X$ set

$$\psi_a := d(\cdot, a) - d_X(a) : M \rightarrow \mathbb{R},$$

a 1-Lipschitz function, negative exactly on the open free ball $B(a, d_X(a))$, with $\psi_a \geq 0$ on X . It is normalised by the reach — $\min \psi_a = -d_X(a)$ at a — rather than by a basepoint; this intrinsic normalisation is what makes the limits below basepoint-free.

Definition 2.3 (Medial limit functions). *A function $\psi : M \rightarrow \mathbb{R}$ is a medial limit function of X if there are $a_k \in M_X^g$ with $\psi_{a_k} \rightarrow \psi$ locally uniformly. It is finite if (a_k) has a bounded subsequence and ideal otherwise (then ψ is a horofunction-type limit: $d(a_k, \cdot) \rightarrow \infty$ locally uniformly while ψ stays finite). Every ψ is 1-Lipschitz, and $\psi \geq 0$ on X (each ψ_{a_k} is), so $\{\psi < 0\}$ is an open set disjoint from X : the free horoball of ψ . For finite ψ along $a_k \rightarrow a$: if $a \in M_X^g$ then $\psi = \psi_a$ is a free-ball function; in general (M_X^g is not closed) $\psi = \psi_a$ with a a focal-degenerate limit of medial centres, or even a point of X itself (medial centres collapsing into a corner of X with $d_X(a_k) \rightarrow 0$), where $\psi_a = d(\cdot, a)$ and the free horoball is empty — still usable, by Lemma 4.1, which never needs the limit centre itself.*

Remark 2.4 (Relation to the horofunction compactification). M proper makes $C(M)$ with the topology of locally uniform convergence a countably compact home for 1-Lipschitz functions normalised at a point (Arzelà–Ascoli). The classical horofunction compactification [22, 14] normalises $d(\cdot, a_k)$ by a basepoint value; our ψ 's differ from it by the scalar $d(o, a_k) - d_X(a_k)$, which is exactly the *offset* bookkeeping of [4, Thm. 3.4]. Working with reach-normalised limits absorbs the offset into the function itself; see Remark 4.6. On a Hadamard manifold every ideal medial limit function is a Busemann function plus a constant (horofunctions of a proper CAT(0) space are Busemann [14, II.8.13]), so ideal medial limit functions are exactly “Busemann function $b_\xi + \text{level } c$ ”.

3 The capture theorem

Theorem 3.1 (Capture or escape). *Let M be complete, X closed, $p \in M \setminus X$, and let Φ be any capture curve from p ; assume $p \notin \Sigma$ (otherwise p is captured at time 0: $a = p$ contains p at depth $d_X(p)$ in its own medial ball). Let*

$$T := \sup\{t \in [0, T_{\max}) : \Phi_s \notin \Sigma \ \forall s \in [0, t]\} \in (0, T_{\max}].$$

Then on $[0, T)$,

$$d_X(\Phi_t) = d_X(p) + t, \quad d(p, \Phi_t) \leq t, \quad \text{hence} \quad d_X(\Phi_t) - d(p, \Phi_t) \geq d_X(p), \quad (2)$$

and exactly one of the following holds.

- (i) (Capture.) $T < T_{\max}$. Then there exist $a \in M_X^g$ arbitrarily close to Φ_T ($a = \Phi_T$ itself if $\Phi_T \in \Sigma$) with

$$d_X(a) - d(p, a) \geq d_X(p) - \varepsilon \quad (\varepsilon > 0 \text{ arbitrary}),$$

so $p \in B(a, d_X(a))$ for such a : $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$, and $T \leq \text{diam}(M) - d_X(p)$.

- (ii) (Escape.) $T = T_{\max}$. Then $T_{\max} = \infty$, the trajectory never meets Σ , $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$, and Φ_t leaves every compact subset of M . This is possible only if M is non-compact.

Proof. The rate-one identity. Fix $t \in [0, T)$; on $[0, t]$ the trajectory avoids Σ , so $U(\Phi_s) = \{u(\Phi_s)\}$ is a singleton for every $s \leq t$ and (1) reads $\Phi'(s) = -u(\Phi_s)$ at a.e. s . The map $h(s) := d_X(\Phi_s)$ is Lipschitz. Let s be any point where both $\Phi'(s)$ exists (with the inclusion valid) and $h'(s)$ exists — almost every $s \in [0, t]$. Write $v = \Phi'(s) = -u(\Phi_s)$, a unit vector. In a chart around Φ_s , differentiability gives $d(\Phi_{s+\sigma}, \exp_{\Phi_s}(\sigma v)) = o(\sigma)$ as $\sigma \downarrow 0$, and d_X is 1-Lipschitz, so

$$h(s + \sigma) = d_X(\exp_{\Phi_s}(\sigma v)) + o(\sigma) = h(s) + \sigma \min_{u \in U(\Phi_s)} \langle v, -u \rangle + o(\sigma) = h(s) + \sigma \langle -u, -u \rangle + o(\sigma),$$

by Lemma 2.2 and single-valuedness. Hence $h'(s) = |u|^2 = 1$ at a.e. $s \in [0, t]$, and integrating, $h(t) = d_X(p) + t$. The bound $d(p, \Phi_t) \leq t$ is the 1-Lipschitz property of the trajectory, and the margin inequality in (2) is their difference. Note that no closedness of Σ , and no structure at focal or conjugate points, was used: a point of $(M \setminus X) \setminus \Sigma$ in the closure of Σ is simply a point where U is single-valued, and the identity passes through it.

Dichotomy. Suppose first $T < T_{\max}$. If $\Phi_T \in \Sigma$, continuity of $d_X(\Phi) - d(p, \Phi)$ and (2) on $[0, T)$ give $d_X(\Phi_T) - d(p, \Phi_T) \geq d_X(p) > 0$, so $a = \Phi_T \in M_X^g$ contains p in its open medial

ball at depth $\geq d_X(p)$. If $\Phi_T \notin \Sigma$, then by definition of T as a supremum, every interval $(T, T + \delta] \subseteq [0, T_{\max})$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$; letting $\delta \downarrow 0$ and using continuity of the margin, $a = \Phi_{s_\delta} \in M_X^g$ satisfies $d_X(a) - d(p, a) \geq d_X(p) - \varepsilon$ for δ small. In both cases $p \in B(a, d_X(a))$ with $a \in M_X^g$, i.e. $p \in \mathcal{R}_X^g$; and $d_X \leq \text{diam}(M)$ bounds T via (2).

Now suppose $T = T_{\max}$: the trajectory never meets Σ , and (2) holds on all of $[0, T_{\max})$. If $T_{\max} < \infty$, then $\Phi([0, T_{\max})) \subseteq \bar{B}(p, T_{\max}) \cap \{d_X \geq d_X(p)\}$, a compact subset of the open set $M \setminus X$ (properness of M); by the extension property of solutions this contradicts maximality. So $T_{\max} = \infty$ and $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$; since d_X is bounded on compact sets, Φ_t leaves every compact set. On compact M , $d_X \leq \text{diam}(M)$ makes this impossible, so case (i) occurs. $\square$

Corollary 3.2 (Total reconstruction on compact manifolds, unconditional). *Let M be a compact Riemannian manifold and $X \subseteq M$ closed, $\emptyset \neq X \neq M$. Then*

$$\boxed{\mathcal{R}_X^g = M \setminus X},$$

and quantitatively: every $p \in M \setminus X$ lies at depth $\geq d_X(p) - \varepsilon$ (any $\varepsilon > 0$) inside a medial ball of M_X^g . No curvature, genericity, analyticity, spread, or cardinality hypothesis is required; in particular $|X| = 1$ is allowed, where the two-point axis has $M_X = \emptyset$ but the two-geodesic axis still reconstructs everything.

Proof. $\mathcal{R}_X^g \subseteq M \setminus X$ since medial balls are free. Conversely, if $p \in M_X^g$ then $p \in B(p, d_X(p)) \subseteq \mathcal{R}_X^g$; otherwise start a capture curve at p (local existence, §2); by Theorem 3.1 escape is impossible on compact M , so p is captured. $\square$

Corollary 3.3 (Two-point axis under spread; strengthening [5, Thm. 6.1]). *Let M be complete with $\text{inj}(M) := \inf_{q \in M} \text{inj}(q) > 0$, and suppose*

$$\sup_{M \setminus X} d_X < \text{inj}(M) \quad \text{— e.g. } X \text{ } \delta\text{-dense with } \delta < \text{inj}(M).$$

Then $\mathcal{R}_X = M \setminus X$ for the two-point axis. Compactness is not needed: it entered [5, Thm. 6.1] only to preclude escape, which the spread hypothesis already does; and the constant improves from the $2\delta < \text{inj}(M)$ of [5, Thm. 6.1] to $\delta < \text{inj}(M)$. In $\mathbb{R}^n$ ($\text{inj} = \infty$) the statement recovers the classical fact that every coarsely dense closed set reconstructs its complement: $\sup d_X < \infty$ implies $\mathcal{R}_X = \mathbb{R}^n \setminus X$.

Proof. Escape is impossible: along an escaping capture curve, $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$ would exceed $\sup d_X$. So every $p \in (M \setminus X) \setminus M_X^g$ is captured (Theorem 3.1(i), within time $\sup d_X - d_X(p)$), giving $a \in M_X^g$ with $p \in B(a, d_X(a))$; for $p \in M_X^g$ take $a = p$. Two distinct minimising geodesics from a with the same endpoint x_0 would make $\exp_a$ non-injective on the open ball of radius $d(a, x_0) = d_X(a) < \text{inj}(M) \leq \text{inj}(a)$, impossible. So the two geodesics of a have distinct endpoints: $a \in M_X$. $\square$

Remark 3.4 (Why stopping at Σ , not flowing through it, is the correct formulation). Two natural strengthenings of Theorem 3.1 are false, and identifying their failure is what shaped the proof. First, *the margin $d_X(\Phi) - d(p, \Phi)$ is not monotone through multi-geodesic arcs*, already in $\mathbb{R}^2$: for $X = \{(0, 0), (0, 2)\}$ and $p = (1, 0.9)$, the Lieutier flow reaches the bisector $\{y = 1\}$ and slides along it, and the margin decreases strictly (from ≈ 1.345 toward the limit 1, the depth of p in the limiting half-plane). Second, even the quadratic margin $d_X^2 - d(p, \cdot)^2$, which is

monotone along unique-foot and bisector arcs in $\mathbb{R}^n$, fails at turning points where the active contact set changes (a three-point configuration in $\mathbb{R}^2$ suffices: $X = \{(0, 0), (0, 2), (10, 5)\}$, with the flow from $p = (1, 0.9)$ turning at $\approx (5.75, 1)$, where the derivative of the quadratic margin is negative). Both phenomena live *on* Σ ; on its complement the rate-one identity (2) is exact. Since reaching Σ is the goal, not an obstacle, the correct move is to stop there — capture is achieved, with the margin intact, *before* the first multi-geodesic time. The failure of closedness of Σ (focal degenerations, e.g. the evolute of an ellipse in $\mathbb{R}^2$, or the conjugate-cusp endpoints of an ellipsoid’s cut segment) is equally harmless: such points are unique-geodesic points, the field $-u$ is continuous at them (Lemma 2.1), and the trajectory passes through with d_X still growing at rate one, the foot sliding continuously.

Remark 3.5 (Relation to the singular gradient flows of Lieutier and Albano–Cannarsa–Nguyen–Sinestrari). For semiconcave d_X one can canonically select the minimal-norm element of G (Lieutier’s generalised gradient [25], in $\mathbb{R}^n$; [8, 26] for the semiconcavity). Two theorems run that canonical flow *into* the singular set: Lieutier’s homotopy equivalence of a bounded domain with its medial axis [25], and its Riemannian counterpart by Albano, Cannarsa, Nguyen and Sinestrari [9], who prove that the singular set of the boundary distance of a bounded domain on a Riemannian manifold is invariant under the generalised gradient flow and deduce the homotopy equivalence of domain and singular set; see Cannarsa–Cheng–Fathi [15] for the current form of the propagation-of-singularities method behind such results. The rate-one growth of d_X off the singular set is classical in that literature; what Theorem 3.1 takes from the flow is different, and in a sense opposite. Our argument analyses the inclusion only *off* Σ , where every selection agrees with the naive one (extend the unique minimising geodesic); it imposes no semiconcave or minimal selection — any relaxed solution serves; and it *terminates* at Σ , where those flows are designed to continue. The deliverable is not a retraction but the quantitative margin $d_X(a) - d(p, a) \geq d_X(p)$ at the first approach — a reconstruction statement with no counterpart in the homotopy results — extracted on an arbitrary complete manifold, for an arbitrary closed X , with the escape alternative live. This is why no analogue of the swallow property past cut points is needed: the trajectory is stopped at first approach, and the reconstruction is read off the margin accumulated strictly before it. The price — capture produces a centre only *near* the first Σ -approach when $\Phi_T \notin \Sigma$ — costs an arbitrarily small ε of depth, invisible to the open-ball statement. Section 6 completes the comparison in the opposite direction: importing their invariance theorem, capture supplies the uniform clock that boundedness of the domain supplied in [25, 9], extending the homotopy equivalence to every complete M with $\sup d_X < \infty$; and the Aubry set of [15] turns out to be exactly the escape class of Theorem 4.4, which locates the failure beyond that bound.

Remark 3.6 (Capture as critical point theory of d_X). The inclusion (1) is also the engine of the critical point theory of distance functions of Grove–Shiohama [24, 17, 23]: $q \in M \setminus X$ is *critical* for d_X iff $0 \in \text{conv } U(q)$, which forces $|U(q)| \geq 2$ — so all critical points of d_X lie on M_X^g — and the isotopy lemma deforms sublevel sets across critical-value-free intervals along precisely such gradient-like curves. In that vocabulary Theorem 3.1 is an isotopy lemma made quantitative and unconditional: it assumes no critical-value-free interval — the trajectory is stopped at its first approach to the (non-closed) set $M_X^g \supseteq \{\text{critical points}\}$, whenever that occurs — and it returns not an isotopy but the margin $d_X(a) - d(p, a) \geq d_X(p)$: a reconstruction statement. The min-type Morse theory of Gershkovich–Rubinstein [21] structures the same objects on nonpositively curved manifolds, where germs of d_X are min-type; and in $\mathbb{R}^n$ the *weak feature size* of Chazal–Lieutier [16] — the smallest critical value of d_X , below which their λ -medial axis carries the

homotopy type of the complement — is the global critical scale whose pointwise, per-trajectory analogue is the capture margin. A Riemannian λ -medial axis built on that margin is a natural project we do not open here.

4 Drift: the swallow, the characterisation, the trichotomy

Capture is complete on compact manifolds, but on a non-compact M a point can be reconstructed by medial balls whose centres go to infinity while its own capture curve escapes. The primal example is $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$: a point p in an open half-plane bounded by the line x_0x_1 , placed beyond x_0 at an obtuse angle, has a capture curve that runs straight to infinity with the single foot x_0 — yet p is swallowed by the bisector balls $B(a_k, r(a_k))$, $a_k \rightarrow \infty$. The limit of those balls is a half-plane, and the correct general home for such limits is the space of medial limit functions of Definition 2.3.

Lemma 4.1 (Swallow, general and offset-free). *Let M be any complete Riemannian manifold, X closed, and ψ a medial limit function of X . Then*

$$\{\psi < 0\} \subseteq \mathcal{R}_X^g.$$

If the witnessing centres lie in M_X , then $\{\psi < 0\} \subseteq \mathcal{R}_X$.

Proof. Let $\psi = \lim \psi_{a_k}$ locally uniformly, $a_k \in M_X^g$, and let $q \in \{\psi < 0\}$. Then $\psi_{a_k}(q) \rightarrow \psi(q) < 0$, so for large k

$$d(q, a_k) - d_X(a_k) = \psi_{a_k}(q) < 0,$$

i.e. $q \in B(a_k, d_X(a_k))$ with $a_k \in M_X^g$: $q \in \mathcal{R}_X^g$. $\square$

No anchoring of feet, no curvature bound, no pinching, no offset hypothesis: the entire content of the swallow lemmas of [4, Lem. 2.2] (and its constant-margin spherical form, [5, Rem. 4.2]) is absorbed into the normalisation of ψ_a by the reach.

Lemma 4.2 (Compactness of reconstructing sequences). *Let $p \in \mathcal{R}_X^g$, and let $a_k \in M_X^g$ be any sequence with $\limsup_k [d_X(a_k) - d(p, a_k)] > 0$ (e.g. a sequence of centres whose balls contain p with margin bounded away from 0; a single ball repeated qualifies). Then a subsequence of (ψ_{a_k}) converges locally uniformly to a medial limit function ψ with $\psi(p) < 0$.*

Proof. The functions ψ_{a_k} are 1-Lipschitz and $\psi_{a_k}(p) = d(p, a_k) - d_X(a_k)$ is bounded: below by $-d_X(p)$ (since $d_X(a_k) \leq d_X(p) + d(p, a_k)$) and above by 0 along the qualifying subsequence. By Arzelà–Ascoli on the proper space M , a further subsequence converges locally uniformly; the limit is a medial limit function, and $\psi(p) = \lim \psi_{a_k}(p) \leq -\limsup [d_X(a_k) - d(p, a_k)] < 0$. $\square$

Theorem 4.3 (Characterisation on an arbitrary complete manifold). *For every complete Riemannian manifold M and every closed $X \subseteq M$, $\emptyset \neq X \neq M$,*

$$\mathcal{R}_X^g = \bigcup \{ \{\psi < 0\} : \psi \text{ a medial limit function of } X \}.$$

The same identity holds for $\mathcal{R}_X$ and M_X in place of $\mathcal{R}_X^g$ and M_X^g . (Every finite medial limit function is itself of the form $\psi_a = d(\cdot, a) - d_X(a)$ with a in the closure of M_X^g taken in M : a focal-degenerate limit of medial centres, or — degenerately — a point of X , where $d_X(a) = 0$ and the sublevel set is empty. Either way its sublevel set is swallowed by the lemma, which never needs the limit centre itself.)

Proof. “ $\supseteq$ ” is Lemma 4.1. “ $\subseteq$ ”: if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$, and ψ_a — a medial limit function via the constant sequence — has $\psi_a(p) < 0$. $\square$

The identity is deliberately economical: its force is not the set equality (the finite members alone give a tautology) but the pair of facts it packages. *First*, the ideal members reconstruct their *entire* open sublevel set — the statement that in [3, 4] required anchoring, pinching, or bisector witnesses, and now holds bare. *Second*, combined with Theorem 3.1 it yields the trichotomy that generalises the hull/drift/ray-escape structure of [4, Thm. 2.1] to arbitrary complete manifolds.

Theorem 4.4 (Trichotomy on a complete manifold). *Let M be complete, X closed, $p \in M \setminus X$. Exactly one of the following holds.*

- (1) (Captured.) *Some capture curve from p approaches M_X^g in finite time. Then $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$.*
- (2) (Swallowed, uncaptured.) *No capture curve from p approaches M_X^g in finite time, but $\psi(p) < 0$ for some medial limit function ψ . Then $p \in \mathcal{R}_X^g$: reconstructed, but only by centres p ’s own ascent cannot find — a remote ball, or a drifting sequence.*
- (3) (Escaping.) *Neither. Then $p \notin \mathcal{R}_X^g$, and every capture curve from p escapes to infinity with $d_X(\Phi_t) = d_X(p) + t$.*

In particular $\mathcal{R}_X^g = \{p : (1) \text{ or } (2)\}$, and on compact M class (1) is all of $M \setminus X$.

Proof. The classes are exclusive and exhaustive by construction. (1) $\Rightarrow p \in \mathcal{R}_X^g$ is Theorem 3.1(i); (2) $\Rightarrow p \in \mathcal{R}_X^g$ is Lemma 4.1. Conversely, if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$ and the finite medial limit function ψ_a has $\psi_a(p) < 0$, so p lies in class (2) whenever it is not in class (1) — never in class (3). Finally, a point not in class (1) has every capture curve uncaptured, hence escaping, by the dichotomy of Theorem 3.1. $\square$

Remark 4.5 (The witnesses of class (2)). Both kinds of witness genuinely occur, already for $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$ and p beyond x_0 in an open half-plane, at an obtuse angle to the bisector: every capture curve from p runs straight to infinity with the single foot x_0 , yet p lies in a single *finite* bisector ball (and also in the ideal half-plane limit of the bisector balls). Class (2) is thus “reconstruction is non-local for p ”: the certificate cannot be found by ascending from p , which is exactly the phenomenon that separates the non-compact theory from the compact one and forces the faithful certificates of [1, 3, 4] to quantify over supporting half-spaces or horoballs rather than over flow lines.

Remark 4.6 (The offset caveat of [4], resolved). Let M be Hadamard. Ideal medial limit functions are exactly $b_\xi + c$ with $\xi \in \partial_\infty M$ and $c \in \mathbb{R}$ (Remark 2.4), normalised so that b_ξ vanishes on the horosphere of the maximal free horoball $\mathcal{H}$ at ξ ; necessarily $c \geq 0$ when X meets every neighbourhood of that horosphere. Call $\mathcal{H}$ *defective at level c* if $b_\xi + c$ is a medial limit function; then Lemma 4.1 reconstructs $\{b_\xi < -c\}$ — the horoball shrunk by c . The “radial offset” $\text{off}_k = d(o, a_k) + b_\xi(a_k)$ of [4, Thm. 3.4] is precisely the discrepancy $\psi_{a_k} - (b_\xi \circ \cdot)$ evaluated along the drift: a drifting sequence with $\text{off}_k \rightarrow c$ is a medial limit at level c . The single unresolved case of [4, Thm. 3.4] — a purely asymptotic drift with non-vanishing offset in an inhomogeneous manifold, where only the deep part $\{\tau_p > \limsup \text{off}_k\}$ was reconstructed — is therefore not a gap in the theorem but the exact content of the level: the drift *is* a level- c certificate and

reconstructs exactly $\{b_\xi < -c\}$, while the full-horoball statement is the assertion that a level-0 certificate exists. Under any of the conditions of [4, Thm. 3.4] (pinching, flat case, a bisector witness on the horosphere — hence in every model of the series) level 0 is achieved and the horoball is reconstructed in full; the hconv-certificate of [4, Thm. 3.3–3.4] detects defectiveness at *some* level unconditionally. The characterisation of the exact level

$$c(\xi) := \inf\{c \geq 0 : b_\xi + c \text{ is a medial limit function}\}$$

from X alone, in an inhomogeneous manifold, is the sharpened form of the remaining question, and it is a question about X and the horosphere geometry, not about the medial axis.

5 Specialisations: the series re-derived

Euclidean space [1]. In $\mathbb{R}^n$ geodesics between two points are unique, so $M_X^g = M_X$ and $\mathcal{R}_X^g = \mathcal{R}_X$. Ideal medial limit functions are affine: $|y - a_k| - r_k$ with $a_k \rightarrow \infty$ subconverges to $y \mapsto -\langle y, u \rangle - c$, whose sublevel set is an open half-space. Theorem 4.3 thus reads: $\mathcal{R}_X$ is the union of the medial balls and of the open half-spaces that are limits of medial balls — and Theorem 6 of [1] is its faithful certificate: such a half-space limit exists iff a supporting half-space has non-convex contact set (Motzkin), and the level is then 0, the drift running over a bisector witness with no offset. The escaping class (3) is exactly Białożył’s unreconstructed set — the rays leaving X with a single foot and no defective facet around them.

Hadamard manifolds (complete, simply connected, variable $\text{sec} \leq 0$) [2, 3, 4]. Here class (2)’s ideal witnesses are Busemann functions at a level (Remark 4.6); the faithful certificate for their existence is the horoconvexity defect $X \cap \partial\mathcal{H} \neq \text{hconv}(X \cap \partial\mathcal{H})$ of [4, Thm. 3.3–3.4], unconditional for the $\subseteq$ direction and achieving level 0 in every model of the series. The hull inclusion $\widehat{X} \setminus X \subseteq \mathcal{R}_X$ [2, Thm. 3.1] populates classes (1)–(2) inside the hull; the trichotomy of [4, Thm. 2.1] (hull/drift/ray-escape) is Theorem 4.4 read through the visual boundary.

The sphere, compact manifolds, space forms [5, 6]. Corollary 3.2 contains: total reconstruction on $\mathbb{S}^n$ ([5, Thm. 4.1], now with the $|X| \geq 2$ hypothesis needed only for the two-point axis — $\mathcal{R}_X^g = \mathbb{S}^n \setminus X$ holds even for a single point, reconstructed from the antipode); total reconstruction on every compact M ([5, Thm. 8.3], [6, Thm. 6.4], caveats deleted); and, new, total $\mathcal{R}_X^g$ on every spherical space form $\mathbb{S}^n/\Gamma$ for *every* closed X — the sharp statement left open in [5, Rem. 7.3], where only the δ -dense case was settled. For the two-point axis the compact theory is unchanged: Corollary 3.3 under spread, and cut-shielding as the exact obstruction in general, with the prolate-ellipsoid transition at eccentricity $c^* = 2.6 \pm 0.1$ [6, Rem. 6.6] — shielded points are precisely class-(1) points of $\mathcal{R}_X^g$ whose capturing centres lie in $M_X^g \setminus M_X$.

Non-compact positive curvature. On a paraboloid of revolution ($\text{sec} > 0$, one flattening end) with X a small circle about the vertex, the vertex is a finite medial centre reconstructing the inner disk (class (1)); every point outside X on the flaring side ascends its meridian forever with a single foot and no medial limit function reaches it: class (3), $\mathcal{R}_X^g \subsetneq M \setminus X$. With X two points, the bisector is unbounded and its escaping medial centres furnish genuinely *ideal* medial limit functions on a $\text{sec} > 0$ manifold — consistent with the trichotomy of [6], since their sublevel sets are unbounded *non-convex* free regions (Busemann functions are concave above zero; it is

the maximal free *convex* regions that are bounded there, not the carriers). Partial-versus-total is decided by compactness, detection flavour by the Hessian sign, exactly as the 2×2 prescribes — now with both rows unconditional.

The flat cylinder: the first certificate beyond the series. The simplest manifold outside every certificate theorem so far is the flat cylinder $C = (\mathbb{R}/L\mathbb{Z}) \times \mathbb{R}$: flat but not simply connected, so neither [1] nor the Hadamard results apply, and non-compact, so Corollary 3.2 is silent about its ends. Its horofunction boundary consists of two points — the compact circle factor washes out of every limit $d(\cdot, a_k) - d(o, a_k)$ — so ideal medial limit functions are $\pm z + \text{const}$, attached to the two ends; write z for the second coordinate and π for the projection to the circle factor.

Proposition 5.1 (End certificate on the flat cylinder). *Let $X \subseteq C$ be closed, $\emptyset \neq X \neq C$, bounded above: $z_+ := \sup_X z < \infty$. The upper contact set $X_+ := X \cap \{z = z_+\}$ is nonempty and compact, and exactly one of the following holds.*

- (i) (Defective end.) X_+ omits a circle position: $\pi(X_+) \neq \mathbb{R}/L\mathbb{Z}$. Then M_X^g meets every slice $\{z = s\}$ for all large s , the free-ball functions of such centres converge locally uniformly (no subsequence needed) to the medial limit function $\psi = z_+ - z$, and

$$\{z > z_+\} \subseteq \mathcal{R}_X^g :$$

the maximal free end-region is reconstructed in full, at level 0.

- (ii) (Capped end.) $X \supseteq (\mathbb{R}/L\mathbb{Z}) \times \{z_+\}$. Then every point above the top circle has a unique minimising geodesic to X (the vertical one), no medial ball meets $\{z > z_+\}$, and $\mathcal{R}_X^g \cap \{z > z_+\} = \emptyset$: the free end-region is not reconstructed at all (in particular no ideal medial limit function $c - z$ exists).

The lower end is symmetric; an end on whose side X is unbounded admits no ideal medial limit function at all ($\psi \geq 0$ on X would fail).

Proof. X closed and $\{z_+ - 1 \leq z \leq z_+\}$ compact make X_+ nonempty and compact; and with $\Theta_\varepsilon := \pi(X \cap \{z \geq z_+ - \varepsilon\})$ the sets Θ_ε are compact, nested, and $\bigcap_{\varepsilon > 0} \Theta_\varepsilon = \pi(X_+) =: \Theta_+$ (a point $(\alpha, a_\varepsilon) \in X$ with $a_\varepsilon \rightarrow z_+$ has limit $(\alpha, z_+) \in X$), so $\Theta_\varepsilon \rightarrow \Theta_+$ in the Hausdorff sense as $\varepsilon \downarrow 0$.

Two elementary facts about $q = (\theta, s)$ with $s > z_+$. First, the horizontal displacement of any minimising geodesic from q to a foot is at most $L/2$ in absolute value — re-wrapping the lift by $\mp L$ strictly shortens otherwise — with equality forcing a second minimising geodesic (both wraps). Hence, comparing with any top point,

$$s - z_+ \leq d_X(q) \leq \sqrt{(s - z_+)^2 + L^2/4} \leq s - z_+ + \frac{L^2}{8(s - z_+)}.$$

Second, every foot (α, a) of q then has $s - a \leq d_X(q)$, i.e. $a \geq z_+ - \varepsilon(s)$ with $\varepsilon(s) := L^2/(8(s - z_+)) \rightarrow 0$: feet of high points are near-top points, and their positions lie in $\Theta_{\varepsilon(s)}$.

(i) Fix α_0 and $\rho > 0$ with the arc of radius ρ about α_0 disjoint from Θ_+ , and s_1 with $\Theta_{\varepsilon(s)}$ contained in the $\rho/2$ -neighbourhood of Θ_+ for $s \geq s_1$. Suppose a slice $\{z = s\}$ with $s \geq s_1$ contained no point of M_X^g . Then every (θ, s) has a unique minimising geodesic, whose

horizontal displacement $\delta(\theta) \in (-L/2, L/2)$ (strict, by uniqueness) is continuous in θ (Lemma 2.1 with single values), and the foot position is $\theta + \delta(\theta) \bmod L$. As θ increases through one period, $\theta + \delta(\theta)$ is continuous with total increment L , so by the intermediate value theorem its values cover *every* circle position — including the forbidden arc about α_0 : a contradiction. So M_X^g meets every slice $\{z = s\}$, $s \geq s_1$; pick a_s there. For (θ, z) in a fixed compact set and s large, the displacement bound gives $s - z \leq d((\theta, z), a_s) \leq s - z + L^2/(8(s - z))$, while $s - z_+ \leq d_X(a_s) \leq s - z_+ + L^2/(8(s - z_+))$; subtracting, $\psi_{a_s} \rightarrow z_+ - z$ locally uniformly. Lemma 4.1 swallows the sublevel set: $\{z > z_+\} \subseteq \mathcal{R}_X^g$.

(ii) If the full top circle lies in X then for $q = (\theta, s)$, $s > z_+$, the vertical segment to $(\theta, z_+) \in X$ has length $s - z_+$, while any geodesic to any $x \in X$ has length $\geq s - z(x) \geq s - z_+$, with equality throughout only for that vertical segment: the minimising geodesic is unique, so $M_X^g \cap \{z > z_+\} = \emptyset$. And no medial ball — centred, therefore, at height $\leq z_+$ — contains a point q above the circle: the minimising geodesic from the centre to q stays in the open ball and crosses $\{z = z_+\} \subseteq X$, contradicting freeness. So $\mathcal{R}_X^g \cap \{z > z_+\} = \emptyset$; and an ideal limit $c - z$ would swallow $\{z > c\}$, so none exists. $\square$

Remark 5.2 (What the cylinder teaches). The dichotomy is decided by X alone — a faithful certificate, the first beyond the geometries of the series. Two degenerations are forced by the *compact* horosphere. First, Białożył’s “non-convex contact set” relaxes to “proper subset of the horosphere circle”: a *single* contact point suffices for the two-geodesic drift, the wrap-around geodesic acting as the second contact. For X a single point this gives $\mathcal{R}_X^g = C \setminus X$ in full — the non-compact counterpart of the $|X| = 1$ clause of Corollary 3.2 — while $M_X = \emptyset$ and $\mathcal{R}_X = \emptyset$ exactly as in $\mathbb{R}^n$: on non-simply-connected manifolds the two axes separate already for singletons. Second, the level is always 0 (a drift centre’s reach differs from its height above z_+ by $O(1/s)$): compact horospheres leave no room for the offset phenomenon, consistent with the expectation that $c(\xi) > 0$, if it occurs at all, requires inhomogeneous infinite-volume horospheres (§8(2)). The homotopy content of the singleton example is recorded in Remark 6.6: the reconstruction is complete while $C \setminus X \not\approx M_X^g$ — the two swallowed escape rays are exactly the Aubry set of d_X , and they carry the fundamental group, which the medial complex of §7 recovers from the two ideal cells.

Remark 5.3 (How much curvature, where). Since the series ranges over constant, pinched, and merely sign-constrained curvature, we record exactly what each layer assumes; “ $\text{sec} \leq 0$ ” etc. always means a *sign* condition on in-general-variable curvature, never constancy, unless said otherwise.

- *The dynamical layer* — capture (Theorem 3.1), swallow (Lemma 4.1), characterisation (Theorem 4.3), trichotomy (Theorem 4.4), and total reconstruction on compact M (Corollary 3.2): *no curvature hypothesis at all*. Variable curvature of either or mixed sign is covered.
- *The Hadamard certificate* [4]: the sign condition $\text{sec} \leq 0$, with variable curvature expressly allowed — the warped products M_f of [4, §5], of non-constant, two-sidedly pinched negative curvature, are its showcase — *together with simple connectedness*. The latter is not cosmetic: it is what identifies horofunctions with Busemann functions of rays (Remark 2.4), and on a non-simply-connected $\text{sec} \leq 0$ manifold the faithful certificate is not established (§8(1)).

- *The knife-edge* [1]: exact flatness *and* simple connectedness, i.e. $M = \mathbb{R}^n$ itself. A flat cylinder has $\text{sec} \equiv 0$ but is not covered by Theorem 6 verbatim: its ideal certificates attach to the two ends, not to a visual sphere of directions — they are instead the dichotomy of Proposition 5.1.
- *Constancy of curvature* is essential in exactly one result of the series: total reconstruction from the two-point axis on $\mathbb{S}^n$ [5, Thm. 4.1]. Its mechanism — every geodesic realises the diameter at its cut point, in every direction at once — is a constant-curvature rigidity, and the conclusion genuinely fails under variable $\text{sec} > 0$ (the prolate-ellipsoid shielding transition at $c^* = 2.6 \pm 0.1$, [6, Rem. 6.6]). The two-*geodesic* counterpart needs no curvature at all (Corollary 3.2).

The updated table. Rows: compactness (axis A). Columns: Busemann–Hessian sign (axis B). Entries: the two-geodesic theory, with the two-point refinement in brackets.

	$\text{sec} \leq 0$	$\text{sec} \equiv 0$	$\text{sec} > 0$
non-compact	partial; hconv defect at level $c(\xi)$ [4]	partial; conv defect (Białożyty [1])	partial (paraboloid) finite + ideal centres
compact	$\text{total } \mathcal{R}_X^g$ [M_X : shielding]	$\text{total } \mathcal{R}_X^g$ [M_X : shielding]	$\text{total } \mathcal{R}_X^g$ [M_X : shielding; = $\mathbb{S}^n$ ok]

The compact row is Corollary 3.2 — unconditional, all curvatures, $|X| \geq 1$. The non-compact row is Theorem 4.3, with the faithful certificates known on the flat and Hadamard columns and open elsewhere (§8). The column heads are sign conditions on variable curvature and govern only the detection flavour; the certificates quoted in the cells carry the stronger hypotheses of their sources — simple connectedness (Hadamard) in the first column, $M = \mathbb{R}^n$ exactly at the knife-edge, with the flat cylinder now carrying its own end certificate (Proposition 5.1) — and constancy of curvature appears only in the bracketed two-point refinement of the $\mathbb{S}^n$ cell (Remark 5.3).

6 The homotopy layer: capture, the Aubry set, and the limits of swallowing

Remark 3.5 separated the capture theorem from the canonical singular gradient flow of Lieutier and Albano–Cannarsa–Nguyen–Sinestrari, whose purpose is homotopy equivalence. This section runs the two together and settles the homotopy question for the completed theory: when the axis carries the homotopy type of the complement, what obstructs it when it does not, and what either answer has to do with the reconstruction $\mathcal{R}_X^g$. Throughout, Φ denotes the *canonical* flow: the unique generalised-gradient trajectory of d_X through each point of $M \setminus X$ — Lieutier’s flow in $\mathbb{R}^n$ [25]; existence and uniqueness [9, Thm. 3.4, Cor. 3.6] in the Euclidean case and [9, §4] on Riemannian domains. The construction is local, its only input being the local semiconcavity of d_X on $M \setminus X$ [8, 26], so it defines Φ for every closed X in every complete M ; uniqueness gives the semigroup property. Canonical trajectories satisfy the inclusion (1), so Theorem 3.1 applies to them verbatim; and they are global in time: d_X is nondecreasing along them, so they stay in $M \setminus X$, and speed ≤ 1 together with completeness of M rules out finite-time breakdown. We import two further facts, both local, from that literature.

- (F1) *Joint continuity*: $(t, p) \mapsto \Phi_t(p)$ is continuous on $[0, \infty) \times (M \setminus X)$. (Local semiconcavity makes the ascending field one-sidedly Lipschitz, and Grönwall gives locally Lipschitz dependence on the initial point; this is the continuity that underlies the homotopy statements of [25, 9].)
- (F2) *Propagation*: Σ is positively invariant under Φ : once a canonical trajectory is singular it stays singular [9, Thms. 3.7 and 4.5]. (Stated there for domains, under curvature bounds; the proof — semiconcavity plus a differential inequality — is local, and a finite time window of a trajectory lives in a compact subset of $M \setminus X$, where the bounds hold. So it applies verbatim here.)

Lemma 6.1 (Capture is permanent; the cut locus is invariant). *Let $p \in M \setminus X$ and suppose the canonical trajectory of p is captured, with first-approach time T as in Theorem 3.1(i). Then*

$$\Phi_t(p) \in \Sigma \quad \text{for every } t > T :$$

the trajectory does not merely approach the axis; it enters it and never leaves. Moreover the cut locus

$$\text{Cut}(X) := \overline{\Sigma} \cap (M \setminus X) \quad (\text{closure in } M)$$

is positively invariant as well.

Proof. If $\Phi_T \in \Sigma$, apply (F2) from time T . If $\Phi_T \notin \Sigma$, then by the definition of T every interval $(T, T + \delta]$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$ (proof of Theorem 3.1); (F2) from time s_δ gives $\Phi_t \in \Sigma$ for all $t \geq s_\delta$, and $\delta \downarrow 0$ covers all of (T, ∞) . For the cut locus: $q \in \text{Cut}(X)$ is a limit of $a_j \in \Sigma$, so by (F1) and (F2), $\Phi_t(q) = \lim_j \Phi_t(a_j) \in \overline{\Sigma}$; and $\Phi_t(q) \in M \setminus X$ always. $\square$

Theorem 6.2 (Uniform capture is a homotopy equivalence). *Let M be complete, X closed, and let $U \subseteq M \setminus X$ be open and positively invariant under the canonical flow, on which capture is uniform: for some $D < \infty$ the trajectory of every $p \in U$ meets Σ by time D . Then $\Phi_D(U) \subseteq \Sigma \cap U$, and the inclusions*

$$\Sigma \cap U \hookrightarrow \text{Cut}(X) \cap U \hookrightarrow U$$

are homotopy equivalences, with Φ_D as homotopy inverse and the flow itself as the homotopy.

Proof. $\Phi_D(p) \in \Sigma$ for $p \in U$ by Lemma 6.1 (the trajectory meets Σ at some $t^* \leq D$ and stays), and $\Phi_D(p) \in U$ by invariance. Set $r := \Phi_D : U \rightarrow \Sigma \cap U$, write $i : \Sigma \cap U \hookrightarrow U$ for the inclusion, and let $H(p, s) := \Phi_{sD}(p)$, jointly continuous by (F1). Then H is a homotopy in U from id_U to $i \circ r$; and H restricted to $(\Sigma \cap U) \times [0, 1]$ takes values in $\Sigma \cap U$ ((F2) and invariance of U), so it is also a homotopy from $\text{id}_{\Sigma \cap U}$ to $r \circ i$. Hence i is a homotopy equivalence. The same two sentences with $\text{Cut}(X) \cap U$ in place of $\Sigma \cap U$ — using the invariance of the cut locus from Lemma 6.1 and $\Phi_D(U) \subseteq \Sigma \cap U \subseteq \text{Cut}(X) \cap U$ — show that $\text{Cut}(X) \cap U \hookrightarrow U$ is a homotopy equivalence, and the inner inclusion follows by two-out-of-three. $\square$

Corollary 6.3 (Bounded distance: topological faithfulness). *Let M be complete, X closed, $\emptyset \neq X \neq M$, and suppose $D := \sup_{M \setminus X} d_X < \infty$. Then escape is impossible, $\mathcal{R}_X^g = M \setminus X$, every component Ω of $M \setminus X$ meets Σ , and*

$$\Sigma \cap \Omega \hookrightarrow \text{Cut}(X) \cap \Omega \hookrightarrow \Omega$$

are homotopy equivalences. In particular:

- (a) on every compact M : $M \setminus X \simeq M_X^g \simeq \text{Cut}(X)$, unconditionally — the closed-set counterpart of the classical retraction of $M \setminus \{x_0\}$ onto $\text{Cut}(x_0)$;
- (b) for every coarsely dense closed set in any complete M — e.g. a periodic $X \subseteq \mathbb{R}^n$ — the complement, an unbounded domain outside the scope of [25, 9], has the homotopy type of its two-geodesic axis and of its cut locus.

With Corollary 3.3 this completes a pairing: $\sup d_X < \text{inj}(M)$ makes the two-point reconstruction metrically complete ($\mathcal{R}_X = M \setminus X$), and $\sup d_X < \infty$ alone makes the two-geodesic picture topologically faithful.

Proof. Escape forces $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$ (Theorem 3.1(ii)), impossible under the bound; so every $p \notin M_X^g$ is captured, hence $p \in \mathcal{R}_X^g$ (Theorem 3.1(i)), and $M_X^g \subseteq \mathcal{R}_X^g$ trivially: $\mathcal{R}_X^g = M \setminus X$. If a canonical trajectory avoided Σ on $[0, \tau]$, the rate-one identity (2) would give $d_X(p) + \tau \leq D$; hence every trajectory meets Σ by time $D - d_X(p) \leq D$, and the meeting point lies in the component of its start. Apply Theorem 6.2 with $U = \Omega$ (open; invariant, since trajectories are continuous and stay in $M \setminus X$). Compact M has $D \leq \text{diam}(M)$; a δ -dense X has $D \leq \delta$. $\square$

Beyond the bound, escape occurs and the statement genuinely changes. The right language is the *Aubry set* of weak KAM theory, and the translation is exact.

Lemma 6.4 (The Aubry set of d_X is the escape class). *Following [15, Def. 1.4], let*

$$\mathcal{A}^*(X) := \{q \in M \setminus X : q = \gamma(t_0) \text{ for some } t_0 > 0 \text{ and some arclength } \gamma : [0, \infty) \rightarrow M \text{ with } d_X(\gamma(t)) = t \ \forall t \geq 0\}$$

the union of the infinite calibrated rays of X . Then:

- (i) $\mathcal{A}^*(X) \cap \Sigma = \emptyset$;
- (ii) $\mathcal{A}^*(X)$ is closed in $M \setminus X$;
- (iii) a point $p \in M \setminus X$ lies in $\mathcal{A}^*(X)$ if and only if its capture curves escape — that is, $\mathcal{A}^*(X)$ is exactly classes (2) $\cup$ (3) of Theorem 4.4, and

$$M \setminus (X \cup \mathcal{A}^*(X)) = \{\text{captured points}\} \supseteq \Sigma,$$

an open set.

Proof. (i) Let $q = \gamma(t_0)$ lie on a calibrated ray. If q had a second minimising geodesic $\sigma \neq \gamma|_{[0, t_0]}$, the curve σ followed by $\gamma|_{[t_0, t]}$ would realise $d_X(\gamma(t)) = t$ for $t > t_0$ with a corner at q , hence could be strictly shortened — contradiction. (The same argument shows that past a point of Σ no minimising geodesic keeps minimising, so calibrated rays avoid Σ outright.)

(ii) Let $q_j = \gamma_j(t_j) \rightarrow q \in M \setminus X$ with γ_j calibrated. Then $t_j = d_X(q_j) \rightarrow d_X(q) > 0$, and the feet $\gamma_j(0)$ eventually lie in the compact set $\bar{B}(q, 2d_X(q)) \cap X$. Calibrated rays are unit-speed minimising geodesics on every initial segment ([15, Rem. 1.5]); by properness of M a subsequence converges locally uniformly to an arclength ray γ , and $d_X(\gamma(t)) = t$ passes to the limit: $q = \gamma(d_X(q)) \in \mathcal{A}^*(X)$.

(iii) All capture curves from $p \notin \Sigma$ agree until their first approach to Σ : off Σ the inclusion (1) is the ODE of the single unit field, and the trajectory is the arclength extension of the minimising geodesic of p (§2). If that trajectory escapes, it never meets Σ and the rate-one

identity holds forever; prepending the minimising segment from the foot of p produces a curve γ with $d_X(\gamma(t)) = t$ for all $t \geq 0$: an infinite calibrated ray through p , so $p \in \mathcal{A}^*(X)$. Conversely let $p = \gamma(t_0) \in \mathcal{A}^*(X)$; by (i) applied at each $\gamma(s)$, $s > 0$, the ray avoids Σ , and since the trajectory off Σ is the extension of the unique minimising geodesic, the trajectory of p is $t \mapsto \gamma(t_0 + t)$: it never meets Σ , and $d_X \rightarrow \infty$ along it — escape. Points of Σ are captured at time 0 and excluded from $\mathcal{A}^*(X)$ by (i); openness of the captured set is (ii). $\square$

Theorem 6.5 (Cannarsa–Cheng–Fathi [15, Thm. 1.6], in the vocabulary of the trichotomy). *For every closed $X \neq \emptyset$ in every complete M , the inclusion*

$$M_X^g \hookrightarrow M \setminus (X \cup \mathcal{A}^*(X)) = \{\text{captured points}\}$$

is a homotopy equivalence: the captured part of the complement always has the homotopy type of the axis — no bound on d_X , no compactness. (Under uniform capture this is Theorem 6.2 with $U = M \setminus X$; in general the uniform clock fails and the statement is exactly [15, Thm. 1.6], translated by Lemma 6.4. Their Theorem 1.1 adds that M_X^g is always locally contractible.)

Remark 6.6 (Homotopy follows capture, not swallowing). The three classes of Theorem 4.4 now have distinct topological meanings. Class (1) carries the homotopy type of M_X^g (Theorem 6.5); the Aubry set $\mathcal{A}^*(X) = (2) \cup (3)$ is what the axis cannot see; and the swallow splits $\mathcal{A}^*(X)$ into its reconstructed part (2) and its free part (3) — a *metric* split that the homotopy type ignores. Both failure modes occur.

- *Escape unswallowed (class (3) active).* X the round circle of radius R in $\mathbb{R}^3$: every axis point has the whole circle as feet, and M_X^g is exactly the axis, contractible. The captured set is the open solid cylinder $\{\rho < R\}$ (each foot-ray there crosses the axis), which deformation-retracts to M_X^g in accordance with Theorem 6.5; the Aubry set is the closed shell $\{\rho \geq R\} \setminus X$, and its unswallowed part — the flange $\{z = 0, \rho > R\}$, the plane of the circle beyond it — is exactly what $\mathcal{R}_X^g = \mathbb{R}^3 \setminus (X \cup \{z = 0, \rho \geq R\})$ misses. The homotopy type $\mathbb{R}^3 \setminus X \simeq S^1 \vee S^2$ is created across that shell and is invisible to the contractible axis.
- *Escape swallowed (class (2) only) — complete reconstruction without homotopy equivalence.* The flat cylinder with X a single point (Proposition 5.1, Remark 5.2): $\mathcal{R}_X^g = C \setminus X$ is complete, yet $C \setminus X \simeq S^1 \vee S^1$ while M_X^g is the opposite generator, contractible. The dictionary locates the leak exactly: $\mathcal{A}^*(X)$ is the punctured generator through the point — two vertical calibrated rays, entirely of class (2), swallowed by the wrap-around balls, whose non-contractibility (radius beyond the injectivity radius) is what lets them reconstruct without transporting homotopy type. The captured set is the open strip $C \setminus \{\text{generator}\}$, contractible, in accordance with Theorem 6.5. Completeness of the reconstruction is a metric statement; it does not imply topological faithfulness.

Under $\sup d_X < \infty$ the Aubry set is empty — d_X is unbounded along a calibrated ray — and the two completions re-fuse: that is Corollary 6.3. Emptiness of $\mathcal{A}^*(X)$ is strictly weaker than bounded distance: inside a parabola in $\mathbb{R}^2$ every normal crosses the symmetry axis before it stops minimising, so $\mathcal{A}^*(X)$ misses the interior component entirely and the (unbounded, unboundedly deep) interior is homotopy equivalent to its medial ray, both contractible. Section 7 closes the leak: what the axis cannot see, the ideal medial limit functions can — every swallowed-uncaptured point lies in an ideal free horoball (Theorem 7.3), and gluing those cells onto the axis restores the homotopy type of $\mathcal{R}_X^g$ (Theorem 7.5). The leak itself is not a curvature phenomenon: it recurs for three points in the plane, where complete reconstruction and a contractible axis coexist with a wedge of three circles (Proposition 7.8).

Two classical instances, and the smooth dual side. The corollary and proposition that close the section read Corollary 6.3 back into classical geometry: uniform capture *is* the spine theory of cusped hyperbolic and crystallographic geometry; and for smooth X the homotopy equivalence is the boundary of a mapping-cylinder structure whose inward dual is the Thom space.

Corollary 6.7 (Spines: cusped hyperbolic manifolds and crystallographic quotients). (a) *Let M be a connected complete hyperbolic manifold of finite volume with cusps, and let X be the union of a system of disjoint closed horoball neighbourhoods of the cusps. Then $\sup d_X < \infty$ and*

$$M \simeq M \setminus X \simeq M_X^g \simeq \text{Cut}(X),$$

where $\text{Cut}(X)$ is the Ford–Voronoi spine of the system — the complex dual to the Epstein–Penner decomposition attached to the same horoballs [20]. The spine theorem of cusped hyperbolic geometry is uniform capture for the horoball contact set.

(b) *Let $M = \mathbb{R}^n/\Gamma$ be a compact flat manifold, $x_0 \in \mathbb{R}^n$, and $X = \{\pi(x_0)\}$ (or any finite set). Then $\text{Cut}(X)$ is the projected wall complex of the Voronoi diagram of the orbit Γx_0 , and $M \setminus X \simeq M_X^g \simeq \text{Cut}(X)$. For the square torus and a single point, $\text{Cut}(X)$ is the union of the two “antipodal” circles — a figure-eight, matching $T^2 \setminus \{x_0\} \simeq S^1 \vee S^1$ — while $M_X = \emptyset$: the singleton has no two-point axis at all, and the two-geodesic axis alone carries the type. The homotopy type that the flat cylinder’s complete reconstruction fails to transmit (Remark 6.6) is carried honestly the moment the manifold is compact.*

Proof. (a) $M \setminus X$ is the open thick-with-tubes part, of compact closure (Margulis; finite volume), so $\sup d_X < \infty$ and Corollary 6.3 applies. For the identification of the axis, lift to the universal cover $\mathbb{H}^n$: the preimage of X is a Γ -invariant family of disjoint closed horoballs. A horoball is convex, so a point outside it has a unique nearest point on it, joined by a unique minimising geodesic; hence two minimising geodesics to X end on two *distinct* horoballs of the lifted family, and Σ is the wall set of the Voronoi decomposition of $\mathbb{H}^n$ determined by the family — whose quotient closure is, by definition, the Ford–Voronoi spine. (The two feet may well project to the *same* point of X in M , so it is M_X^g , not M_X , that carries the spine — the two-geodesic axis is again the right object.) Finally $M \simeq M \setminus X$: the closed cusp neighbourhoods retract onto their horospherical boundaries, and the closed thick part onto the open one along a collar. (b) One level down: the lift of X is the orbit Γx_0 (a Delone set; finitely many orbits for a finite set), Σ projects from the walls of its classical Voronoi diagram, and d_X is bounded by the covering radius, so Corollary 6.3 applies. On $\mathbb{R}^2/\mathbb{Z}^2$ with $x_0 = 0$: the Voronoi cell is the unit square centred at 0, its boundary projects to the two circles $\{x = \frac{1}{2}\} \cup \{y = \frac{1}{2}\}$, meeting only at $(\frac{1}{2}, \frac{1}{2})$: a wedge of two circles. $\square$

Proposition 6.8 (Smooth contact set: the captured part is a mapping cylinder). *Let X be a closed embedded C^∞ submanifold of the complete manifold M , $S(\nu)$ its unit normal bundle,*

$$\rho(v) = \sup\{t > 0 : d_X(\exp(tv)) = t\} \in (0, \infty]$$

the cut time, $S_{\text{fin}} = \{\rho < \infty\}$, and $\sigma : S_{\text{fin}} \rightarrow M \setminus X$, $\sigma(v) = \exp(\rho(v)v)$, the swallow map, whose image $\text{Cu}(X) := \sigma(S_{\text{fin}})$ is the classical cut locus. Then $\mathcal{A}^(X) = \{\exp(tv) : \rho(v) = \infty, t > 0\}$, and the normal exponential descends to a homeomorphism*

$$\text{Cyl}(\sigma) := (S_{\text{fin}} \times (0, 1] \sqcup \text{Cu}(X)) / [(v, 1) \sim \sigma(v)] \xrightarrow{\sim} M \setminus (X \cup \mathcal{A}^*(X)), \quad (v, t) \mapsto \exp(t\rho(v)v) :$$

for smooth X , with no further hypothesis, the captured part of the complement is the mapping cylinder of the swallow map, and it strong deformation retracts onto $\text{Cu}(X)$. In particular $\Sigma \subseteq \text{Cut}(X) \subseteq \text{Cu}(X)$, the inclusion $\Sigma \hookrightarrow \text{Cu}(X)$ is a homotopy equivalence for every smooth closed X (Theorem 6.5 and two-out-of-three) — the equivalence of [15] sharpened, on the smooth case, to a retraction onto the full cut locus — and $\text{Cut}(X) = \text{Cu}(X)$ when $\dim X = 0$, by Bishop's theorem [12]. If $\mathcal{A}^*(X) = \emptyset$ — equivalently $\rho < \infty$ everywhere, in particular whenever $\sup d_X < \infty$ — then $M \setminus X$ itself is the cylinder, and under uniform capture the whole chain $\Sigma \hookrightarrow \text{Cut}(X) \hookrightarrow \text{Cu}(X) \hookrightarrow M \setminus X$ consists of homotopy equivalences.

Proof. Minimising geodesics from a point to X arrive orthogonally (first variation), so every $p \in M \setminus X$ is $\exp(d_X(p)v)$ for some $v \in S(\nu)$ with $d_X(p) \leq \rho(v)$; the calibration interval $\{t : d_X(\exp(tv)) = t\}$ is $(0, \rho(v)]$ (closed at ρ by continuity); and $\rho(v) = \infty$ makes $t \mapsto \exp(tv)$ an infinite calibrated ray, so Lemma 6.4 identifies $\mathcal{A}^*(X)$ with the union of the open rays $\{\exp(tv) : t > 0\}$, $\rho(v) = \infty$. Positivity of ρ is the tubular neighbourhood theorem; continuity of $\rho : S(\nu) \rightarrow (0, \infty]$ is the classical continuity of the cut time, proved for submanifolds as for points (see [10] and the references there, and [30, III.4] for a point).

Write $F(v, t) = \exp(t\rho(v)v)$ on $S_{\text{fin}} \times (0, 1]$, continuous since ρ is finite and continuous on the open set S_{fin} . *Image and surjectivity:* a captured p outside Σ has a unique foot direction v , and $\rho(v) = \infty$ would put $p \in \mathcal{A}^*(X)$; so $p = F(v, d_X(p)/\rho(v))$. A point $p \in \Sigma$ has each of its directions v stopped at p — past a point of Σ no minimising geodesic keeps minimising (Lemma 6.4(i)) — so $\rho(v) = d_X(p)$ and $p = \sigma(v)$: thus $\Sigma \subseteq \text{Cu}(X)$ and F maps onto the captured part; and the image avoids $\mathcal{A}^*(X)$, since a point of $\mathcal{A}^*(X) \cap \text{im } F$ would carry two distinct minimising directions and lie in Σ , while $\mathcal{A}^*(X) \cap \Sigma = \emptyset$. *Injectivity off $t = 1$:* if $p = F(v, t)$ with $t < 1$ then $d_X(p) < \rho(v)$; a second representation would again put $p \in \Sigma$ and force $d_X(p) = \rho$, a contradiction — the same argument shows $F(v, t) \notin \text{Cu}(X)$ for $t < 1$. Hence F descends to a continuous bijection $\bar{F} : \text{Cyl}(\sigma) \rightarrow M \setminus (X \cup \mathcal{A}^*(X))$. *Properness:* let K be a compact subset of the captured part, $0 < \varepsilon \leq d_X \leq C$ on K . Feet of points of K lie in the compact $X_K = \{x \in X : \text{dist}(x, K) \leq C\}$; if $v_k \in S_{\text{fin}}|_{X_K}$ contribute to $\bar{F}^{-1}(K)$ and $v_k \rightarrow v$ with $\rho(v) = \infty$, then the limit point $\lim F(v_k, t_k) = \exp(sv)$ with $s = \lim t_k \rho(v_k) \in [\varepsilon, C]$ would lie in $\mathcal{A}^*(X) \cap K$ — impossible; so the directions stay in a compact subset of S_{fin} on which $\rho \leq R$, the F -preimage of K is a closed subset of that compactum $\times [\varepsilon/R, 1]$, and $\text{Cu}(X) \cap K$ is compact because $\text{Cu}(X)$ is closed in the captured part (if $\sigma(v_k) \rightarrow q$ captured, feet confined give $v_k \rightarrow v$ after a subsequence, $\rho(v_k) = d_X(\sigma(v_k)) \rightarrow d_X(q)$ finite, so by continuity $\rho(v) = d_X(q)$ and $q = \sigma(v)$). A proper continuous bijection onto a locally compact Hausdorff space is a homeomorphism; the cylinder coordinate $H_s(v, t) = (v, t + s(1 - t))$, the identity on $\text{Cu}(X)$, is a strong deformation retraction. Finally $\text{Cut}(X) \subseteq \text{Cu}(X)$: if $a_k \in \Sigma \rightarrow q \in M \setminus X$, pick feet x_k of a_k with normal directions $w_k \in S(\nu)$; points of Σ sit at their own cut time (the corner argument above), so $\rho(w_k) = d_X(a_k) \rightarrow d_X(q)$; the feet are confined, so after a subsequence $w_k \rightarrow w$, and continuity of ρ gives $\rho(w) = d_X(q) < \infty$ and $q = \sigma(w) \in \text{Cu}(X)$. (In particular $\bar{\Sigma}$ avoids $\mathcal{A}^*(X)$ for smooth X — a statement we do not claim for general closed X , where no continuous cut time is available: the merging of directions that keeps M_X^g from being closed is not, by itself, excluded from happening along a calibrated ray.) For $\dim X = 0$, Bishop's theorem [12] applied at the strictly nearest point gives $\text{Cu}(X) \subseteq \bar{\Sigma}$, whence $\text{Cut} = \text{Cu}$. The final chain is Theorem 6.2 plus the retraction just built plus two-out-of-three. $\square$

Remark 6.9 (The two flows, and the Thom space). Proposition 6.8 is the outward half of a picture whose inward half is due to Basu and Prasad [10] (see also the survey [29]): for a closed

embedded submanifold X of a complete M , the square d_X^2 is Morse–Bott on $M \setminus \text{Cu}(X)$, its gradient flow deformation-retracts $M \setminus \text{Cu}(X)$ onto X , and for closed M the collapse $M/\text{Cu}(X)$ is the Thom space of the normal bundle of X . The same normal geodesics thus carry three retractions: inward to X below the cut time [10]; outward to $\text{Cu}(X)$ across it (Proposition 6.8); and — the only one that survives when X is an arbitrary closed set — the canonical flow into the axis, which turns at Σ instead of stopping (Theorem 6.2). For smooth X the three compute the homotopy type of every space in sight from medial data: $M \setminus (X \cup \mathcal{A}^*) \simeq \text{Cu}(X)$, $M \setminus \text{Cu}(X) \simeq X$, $M/\text{Cu}(X) \cong \text{Th}(\nu)$. On the flat cylinder with $|X| = 1$ (Remark 6.6) the picture is visible whole: $\rho = \infty$ exactly at the two vertical directions — the Aubry rays — and the captured strip is the mapping cylinder of $\sigma : S^1 \setminus \{\pm\} \rightarrow \{\text{opposite generator}\}$. Two further notes. (i) *Stability*: $\text{Cu}(X)$ is Hausdorff-stable under C^2 perturbations of the metric and of the submanifold [11], so the cylinder structure is a robust model, in the spirit of the λ -medial axis [16]. (ii) *Retraction versus equivalence*: for arbitrary closed X , even Lieutier’s Euclidean equivalence [25] is, to our knowledge, not known to improve to a deformation retraction — the obstruction is the continuity of a canonical centre selection — so the smooth case, where the retraction is explicit and strong, is genuinely special.

7 The medial complex: the ideal cells close the leak

Remark 6.6 located the failure of topological faithfulness precisely: the axis carries the homotopy type of the captured class (1), and what it cannot see is the Aubry set $(2) \cup (3)$, which the swallow splits by a metric criterion the homotopy type ignores. The flat cylinder suggested the repair — glue the free horoballs of the ideal medial limit functions onto the captured part — and an earlier draft of this note could only conjecture that the resulting *medial complex* always carries the homotopy type of $\mathcal{R}_X^g$. This section proves it, on every complete manifold, for every closed X , with no hypothesis. The entire content is a covering theorem: *the finite layer does not leak*. Every point of class (2) — swallowed, but invisible to its own ascent — lies in an *ideal* free horoball, at depth no smaller than its depth in any finite medial ball. Granting that, the captured set and the ideal free horoballs form an open cover of $\mathcal{R}_X^g$, the homotopy colimit of its diagram is $\mathcal{R}_X^g$ by the gluing calculus of Dugger–Isaksen [19], and the captured vertex is identified with M_X^g by Theorem 6.5.

Throughout, $W := M \setminus (X \cup \mathcal{A}^*(X))$ denotes the captured set, open by Lemma 6.4, and for $p \in M \setminus X$ and $a \in M$ we write

$$u_p(a) := d_X(a) - d(a, p),$$

the *margin of p at a* : $u_p(a) > 0$ with $a \in M_X^g$ says exactly that the medial ball of a swallows p at depth $u_p(a)$. As a difference of 1-Lipschitz functions, u_p is 2-Lipschitz in a (and in p).

Lemma 7.1 (Margin transport). *Let $x \in M \setminus X$, let Φ be a capture curve from x , and let $T = \sup\{t : \Phi_s \notin \Sigma \ \forall s \leq t\}$ be its first approach to the axis as in Theorem 3.1. Then for every $p \in M$ the margin $t \mapsto u_p(\Phi_t)$ is nondecreasing on $[0, T]$ and 2-Lipschitz on all of $[0, T_{\max})$:*

$$u_p(\Phi_t) \geq u_p(x) \quad (t \leq T), \quad |u_p(\Phi_t) - u_p(\Phi_s)| \leq 2|t - s|.$$

Moreover, on any interval $[0, t]$ on which Φ avoids Σ , the curve is the arclength extension of the minimising geodesic of x : capture curves are unique, and are calibrated geodesics, until their first approach.

Proof. On $[0, T]$ the rate-one identity (2) gives $d_X(\Phi_t) = d_X(x) + t$, while $d(\Phi_t, p) \leq d(x, p) + t$ since Φ is 1-Lipschitz; subtracting, $u_p(\Phi_t) \geq u_p(x)$, and continuity extends this to $t = T$. The Lipschitz bound is the 2-Lipschitz property of u_p along a 1-Lipschitz curve. For uniqueness: on an avoidance interval, $d_X(\Phi_t) = d_X(x) + t$ and $d(x, \Phi_t) \leq t$ force $d(x, \Phi_t) = t$ (as $d_X(\Phi_t) \leq d_X(x) + d(x, \Phi_t)$), so $\Phi|_{[0, t]}$ has length $t = d(x, \Phi_t)$ and is a minimising geodesic; the concatenation $[x \rightarrow \Phi_t]$ has length $d_X(x) + t = d_X(\Phi_t)$, hence minimises, hence has no corner at x : $\Phi|_{[0, t]}$ is the geodesic extension of the minimising geodesic of x , as in the proof of Lemma 6.4(iii). $\square$

Lemma 7.2 (Cut rays: grazing trajectories ride the cut locus to infinity). *Let $z \in \mathcal{A}^*(X) \cap \text{Cut}(X)$. Then the (unique) capture curve Φ from z — the shift of the calibrated ray of z , by Lemma 6.4(iii) — satisfies*

$$\Phi_t(z) \in \mathcal{A}^*(X) \cap \text{Cut}(X), \quad d_X(\Phi_t(z)) = d_X(z) + t, \quad \text{for every } t \geq 0 :$$

an escaping ray that touches the closure of the axis stays in it forever.

Proof. By Lemma 6.4(iii) and its proof, every capture curve from z is the arclength shift of the calibrated ray of z ; it never meets Σ , and the rate-one identity holds for all time. In particular the canonical trajectory of §6 is this ray shift. By Lemma 6.1 the cut locus $\text{Cut}(X) = \bar{\Sigma} \cap (M \setminus X)$ is positively invariant under the canonical flow, so $\Phi_t(z) \in \text{Cut}(X)$; and forward points of a calibrated ray lie in $\mathcal{A}^*(X)$ by definition. $\square$

Whether $\mathcal{A}^*(X) \cap \text{Cut}(X)$ can be nonempty for some closed X — merging medial directions accumulating along an escaping calibrated ray — is the regularity question already flagged in Proposition 6.8 (it is empty for smooth X , by continuity of the cut time; the structure theory of [31, 27] is the natural tool for the general case). The point of Lemma 7.2 is that the theorem below does not need to know: a grazing ray is not an obstruction but a certificate, since it drags medial centres with it to infinity.

Theorem 7.3 (Ideal swallow: the finite layer does not leak). *Let M be complete, X closed, and let $p \in \mathcal{A}^*(X) \cap \mathcal{R}_X^g$ — a swallowed, uncaptured point, class (2) of Theorem 4.4. Then for every finite witness $a_0 \in M_X^g$ with $m_0 := u_p(a_0) > 0$ there is an ideal medial limit function ψ with*

$$\psi(p) \leq -m_0 < 0 :$$

p lies in an ideal free horoball, at depth at least its depth in any finite medial ball.

Proof. Let γ be the calibrated ray of p , $p = \gamma(t_0)$ with $t_0 = d_X(p)$, and let $\eta : [0, L] \rightarrow M$, $L = d(a_0, p)$, be a minimising geodesic from a_0 to p ; it stays in the free ball $B(a_0, d_X(a_0)) \subseteq M \setminus X$. Concatenate:

$$\sigma(s) := \eta(s) \quad (0 \leq s \leq L), \quad \sigma(L+t) := \gamma(t_0+t) \quad (t \geq 0).$$

Along η , $d(\eta(s), p) = L - s$ and $d_X(\eta(s)) \geq d_X(a_0) - s$ give $u_p(\sigma(s)) \geq m_0$; along the ray, $d_X(\gamma(t)) = t$ and $d(\gamma(t), p) = t - t_0$ give $u_p = t_0 \geq m_0$ (the second inequality because $u_p(\sigma(\cdot)) \geq m_0$ persists to $s = L$ by continuity, where $u_p(p) = d_X(p) = t_0$). So

$$u_p(\sigma(s)) \geq m_0 \quad \text{for all } s \geq 0. \quad (3)$$

The endpoint data are opposite: $\sigma(0) = a_0 \in \Sigma$ is captured (at time 0), while $p = \sigma(L) \in \mathcal{A}^*(X)$; the captured set is open and $\mathcal{A}^*(X)$ is closed in $M \setminus X$ (Lemma 6.4), so

$$s^* := \inf \sigma^{-1}(\mathcal{A}^*(X)) \in (0, L], \quad x_* := \sigma(s^*) \in \mathcal{A}^*(X), \quad u_p(x_*) \geq m_0,$$

and $\sigma(s)$ is captured for every $s < s^*$.

Case A (the transition grazes the cut locus): suppose there are $s_j \uparrow s^*$ with $\sigma(s_j) \in \bar{\Sigma}$. Then $x_* \in \text{Cut}(X) \cap \mathcal{A}^*(X)$. By Lemma 7.2 the points $w_t := \Phi_t(x_*)$ lie in $\text{Cut}(X)$ with $d_X(w_t) = d_X(x_*) + t \rightarrow \infty$, and by Lemma 7.1 (the trajectory never meets Σ , so $T = \infty$) $u_p(w_t) \geq u_p(x_*) \geq m_0$. Pick $a_t \in \Sigma$ with $d(a_t, w_t) \leq \min(1, 1/t)$: then $a_t \in M_X^g$,

$$u_p(a_t) \geq m_0 - 2/t, \quad d_X(a_t) \geq d_X(x_*) + t - 1/t \rightarrow \infty.$$

Since d_X is bounded on bounded sets, (a_t) has no bounded subsequence. By Lemma 4.2 a subsequence of (ψ_{a_t}) converges locally uniformly to a medial limit function ψ with $\psi(p) = \lim(-u_p(a_t)) \leq -m_0$, and ψ is ideal. Done.

Case B (the transition is clean): otherwise there is $s_1 < s^*$ with $\sigma(s) \notin \bar{\Sigma}$ for $s \in [s_1, s^*)$. Take $s_j \uparrow s^*$ in $[s_1, s^*)$ and set $x_j := \sigma(s_j)$: captured, not in $\bar{\Sigma}$. Each x_j has a capture curve Φ^j with finite first approach T_j , and by the dichotomy proof of Theorem 3.1 there are times $t_j \in [T_j, T_j + 1/j]$ with $b_j := \Phi_{t_j}^j \in \Sigma = M_X^g$. By Lemma 7.1,

$$u_p(b_j) \geq u_p(x_j) - 2/j \geq m_0 - 2/j, \quad d_X(b_j) \geq d_X(x_j) + T_j - 1/j.$$

If (T_j) is unbounded, pass to a subsequence with $T_j \rightarrow \infty$: then $d_X(b_j) \rightarrow \infty$, and exactly as in Case A, Lemma 4.2 produces an ideal ψ with $\psi(p) \leq -m_0$. If (T_j) is bounded, pass to a subsequence $T_j \rightarrow \bar{T} < \infty$. On $[0, T_j)$ each Φ^j avoids Σ , hence (Lemma 7.1) is the geodesic extension of the minimising geodesic of x_j . Since $x_* \notin \Sigma$ ($\mathcal{A}^*(X) \cap \Sigma = \emptyset$), the direction field U is single-valued and continuous at x_* (Lemma 2.1), so the extensions converge locally uniformly to the extension of the minimising geodesic of x_* — which is the calibrated ray of x_* — and, as $d(b_j, \Phi_{T_j}^j) \leq 1/j$,

$$b_j \longrightarrow z := \Phi_{\bar{T}}(x_*) \in \mathcal{A}^*(X) \cap \bar{\Sigma}, \quad u_p(z) = \lim u_p(b_j) \geq m_0 :$$

a grazing point with full margin. Case A applied at z (its hypothesis is exactly $z \in \mathcal{A}^*(X) \cap \text{Cut}(X)$) finishes the proof. $\square$

Corollary 7.4 (Ideal decomposition of the reconstruction). *For every complete M and closed X ,*

$$\mathcal{R}_X^g = W \cup \bigcup \{ \{ \psi < 0 \} : \psi \text{ an ideal medial limit function of } X \}$$

— the captured set plus the ideal layer; the finite medial balls are needed only where capture already operates. On the Aubry set the trichotomy is decided by the ideal functions alone:

$$\text{class (2)} = \mathcal{A}^*(X) \cap \bigcup_{\psi \text{ ideal}} \{ \psi < 0 \}, \quad \text{class (3)} = \mathcal{A}^*(X) \setminus \bigcup_{\psi \text{ ideal}} \{ \psi < 0 \} :$$

a point escapes unswallowed if and only if it is Aubry and no ideal medial limit function is negative on it. In $\mathbb{R}^n$ this sharpens Theorem 6 of [1]: the half-space layer alone covers the entire drift-swallowed set.

Proof. $W \subseteq \mathcal{R}_X^g$ is Theorem 4.4(1), the horoballs are swallowed by Lemma 4.1, and conversely $\mathcal{R}_X^g \setminus W = \mathcal{A}^*(X) \cap \mathcal{R}_X^g$ is covered by Theorem 7.3. The class identities follow since $\mathcal{A}^*(X) \cap W = \emptyset$. $\square$

Theorem 7.5 (The medial complex carries the reconstruction). *Let M be complete, X closed, and let Ψ be any family of ideal medial limit functions whose free horoballs cover $\mathcal{A}^*(X) \cap \mathcal{R}_X^g$ (by Theorem 7.3 the family of all of them qualifies). Then*

$$\mathcal{U} := \{W\} \cup \{\{\psi < 0\} : \psi \in \Psi\}$$

is an open cover of $\mathcal{R}_X^g$, and the natural map from the homotopy colimit of its Čech diagram (the members and their finite intersections) to $\mathcal{R}_X^g$ is a weak equivalence [19] — a homotopy equivalence, since these spaces are ANRs. Since $M_X^g \hookrightarrow W$ is a homotopy equivalence (Theorem 6.5), the medial complex — M_X^g together with the ideal free horoballs, glued along the pattern of their overlaps with the captured set and with one another — carries the homotopy type of $\mathcal{R}_X^g$; and of $M \setminus X$ whenever the reconstruction is complete. The conjecture of the earlier draft is true, unconditionally.

Proof. Covering is Corollary 7.4 together with $\{\psi < 0\} \subseteq \mathcal{R}_X^g$ (Lemma 4.1) and $W \subseteq \mathcal{R}_X^g$; all members are open. The homotopy-colimit statement for an arbitrary open cover is [19]; open subsets of manifolds and their homotopy colimits have CW type, so the weak equivalence upgrades by Whitehead. The last sentence substitutes the CCF equivalence at the captured vertex. $\square$

Corollary 7.6 (Faithfulness without an ideal layer). *If X admits no ideal medial limit function — e.g. whenever M_X^g is bounded, so that no sequence of medial centres escapes — then class (2) is empty, $\mathcal{R}_X^g = W$, and*

$$\mathcal{R}_X^g \simeq M_X^g :$$

the two-geodesic axis carries the homotopy type of everything it reconstructs. This extends the faithfulness regime of Corollary 6.3 beyond bounded distance: d_X may be unbounded and the Aubry set nonempty, as on the paraboloid of §5 (M_X^g the vertex, $\mathcal{R}_X^g$ the inner disk, both contractible, the entire flaring end escaping), or for the round circle in $\mathbb{R}^2$ (M_X the centre, $\mathcal{R}_X$ the open disk).

Proof. Class (2) = $\mathcal{A}^*(X) \cap \bigcup_{\text{ideal}} \{\psi < 0\} = \emptyset$ by Corollary 7.4; so $\mathcal{R}_X^g = W \simeq M_X^g$ by Theorem 6.5. $\square$

The examples, reassembled. On the flat cylinder with X a point (Remark 6.6): W is the open strip, M_X^g the opposite generator, and the two ideal cells are the open half-cylinders $\{\pm z > 0\}$, each a homotopy circle, with $\{z > 0\} \cap \{z < 0\} = \emptyset$ and each W -overlap an open square, contractible. The complex is the double mapping cylinder

$$S^1 \longleftarrow * \longrightarrow * \longleftarrow * \longrightarrow S^1 \simeq S^1 \vee S^1,$$

the figure-eight: exactly $C \setminus X$. The leak of Remark 6.6 is not an anomaly of the theory but the visible size of its ideal cells. For the round circle in $\mathbb{R}^3$: $W = \{\rho < R\}$ is the captured solid cylinder, the two ideal cells are the half-spaces $\{\pm z > 0\}$, all three contractible with contractible overlaps, so the complex is contractible — and indeed $\mathcal{R}_X^g = \mathbb{R}^3 \setminus (X \cup \{z = 0, \rho \geq R\})$ is star-shaped about the centre. Here $M \setminus X \simeq S^1 \vee S^2$ differs from $\mathcal{R}_X^g$: the complex is faithful to the *reconstruction*, and the discrepancy is exactly class (3), the unswallowed flange. For two points in $\mathbb{R}^2$: strip plus two half-planes, all contractible over contractible overlaps; $\mathcal{R}_X^g$, the plane minus two closed rays, is contractible. In all three, finitely many maximal cells suffice as Ψ ; Theorem 7.5 is indifferent to the choice.

Remark 7.7 (What the complex decides, and what it does not). (a) *Levels*. Any covering family Ψ computes the same homotopy type, so the complex never needs the optimal level: the level function $c(\xi)$ of Remark 4.6 (item (2) of §8) is untouched, as is the detection question — *which* ideal functions exist is still read from M_X^g , not from X (items (1)–(2)). (b) *The flat no-leak conjecture, reduced — and refuted*. In $\mathbb{R}^n$ every ideal medial limit function is affine and its cell $H = \{\psi < 0\}$ is an open half-space avoiding X ; along a calibrated ray $\gamma(t) = x_\gamma + tu$ the affine function $t \mapsto \psi(\gamma(t))$ starts nonnegative ($\psi \geq 0$ on X) and is monotone, so a ray that enters H never leaves: $\mathcal{A}^*(X) \cap H$ is a bundle of inward half-lines. The conjecture that $\mathcal{R}_X = \mathbb{R}^n \setminus X$ forces $\mathbb{R}^n \setminus X \simeq M_X$ is therefore equivalent, by Theorem 7.5, to the statement that these attachments are trivial: $H \cap W \hookrightarrow H$ and its finite-intersection versions are weak equivalences. For a single cell both spaces are contractible candidates — H is convex, and $H \cap W$ is H minus the half-line bundle — and on the acute triangle below the single-cell attachments are indeed trivial. The conjecture fails anyway: Proposition 7.8 exhibits double overlaps $H' \cap H''$ that sink entirely into the Aubry set, and three points in the plane already leak all of their homotopy. On a Hadamard manifold the corner mechanism is throttled by the thinness of horoballs — a calibrated ray not asymptotic to the cell’s ideal point eventually exits it, so three points in $\mathbb{H}^2$ do not even reconstruct completely — and which complete Hadamard reconstructions leak is open. (c) *Regularity*. When the cover is good — all members and intersections contractible, as below the convexity radius, or for all of $\mathbb{R}^n$ if (b) holds — the nerve theorem [13] replaces the homotopy colimit by a simplicial complex, and Damon’s skeletal calculus [18] realises exactly this reconstruction-from-medial-data programme for compact generic regions in $\mathbb{R}^3$; the local-tree and delta-convex structure of [31, 27] is what a CW model of the complex would be built on. (d) *Where the classes live*. The escape flow $E_t(x) := \Phi_t(x)$ restricts to $\mathcal{A}^*(X)$ (unique rays, no branching), is continuous there by Lemma 2.1, and E_R is a homotopy inverse to $\mathcal{A}^*(X) \cap \{d_X \geq R\} \hookrightarrow \mathcal{A}^*(X)$ for every R : classes (2) $\cup$ (3) are carried entirely at infinity, which is why only ideal objects can see them — the geometric content of Theorem 7.3.

Proposition 7.8 (The flat leak: three points refute the no-leak conjecture). *Let $X \subset \mathbb{R}^2$ be three affinely independent points. Then the reconstruction is complete, and*

$$\mathcal{R}_X = \mathbb{R}^2 \setminus X \simeq S^1 \vee S^1 \vee S^1, \quad \text{while } M_X \text{ is the Voronoi tripod, contractible.}$$

Complete reconstruction does not transport homotopy type even in $\mathbb{R}^2$, even for $|X| = 3$: the flat no-leak conjecture of Remark 7.7(b) is false.

Proof. Write e for a generic edge of the triangle, ℓ_e for its line, n_e for its outward unit normal and m_e for its midpoint. M_X is the Voronoi skeleton of three points — the circumcentre with its three outward bisector rays — a tripod.

Ideal cells. Along the outward bisector ray the centres $a_t = m_e + tn_e$ have exactly the endpoints of e as feet and $d_X(a_t) = \sqrt{t^2 + h^2}$, $2h = |e|$; hence $d(x, a_t) - d_X(a_t) \rightarrow \psi_e(x) := \langle m_e - x, n_e \rangle$ locally uniformly, an *ideal* medial limit function whose cell $H_e = \{\psi_e < 0\}$ is the open half-plane beyond ℓ_e .

The Aubry set. In $\mathbb{R}^n$ the unit-speed ray $t \mapsto p + tv$ from $p \in X$ is calibrated iff $\bigcup_{t>0} B(p + tv, t) = \{x : \langle x - p, v \rangle > 0\}$ avoids X , i.e. iff v lies in the normal cone N_p of $\text{conv } X$ at p ; and every Aubry point lies on such a ray through its foot (Lemma 6.4). Hence $\mathcal{A}^*(X) = \bigcup_i (p_i + N_{p_i})$, three closed corner cones.

Complete reconstruction. Fix a vertex p , let n', n'' be the normals of its two edges, so $N_p = \text{cone}(n', n'')$ and $p \in \ell' \cap \ell''$. If $v = an' + bn''$ with $a, b \geq 0$ had $\langle v, n' \rangle \leq 0$ and $\langle v, n'' \rangle \leq 0$,

then $|v|^2 = a \langle n', v \rangle + b \langle n'', v \rangle \leq 0$, so $v = 0$. Thus every $v \in N_p \setminus \{0\}$ has, say, $\langle v, n' \rangle > 0$, and since $p \in \ell'$,

$$\psi'(p + tv) = -t \langle v, n' \rangle < 0 \quad (t > 0) :$$

the ray lies in the adjacent cell from the instant it leaves p . So $\mathcal{A}^*(X) \subseteq H_1 \cup H_2 \cup H_3$, and Corollary 7.4 gives $\mathcal{R}_X^g \supseteq W \cup \mathcal{A}^*(X) = \mathbb{R}^2 \setminus X$, i.e. $\mathcal{R}_X = \mathcal{R}_X^g = \mathbb{R}^2 \setminus X$ ($M_X = M_X^g$ in $\mathbb{R}^n$).

Finally $\mathbb{R}^2$ minus three points is a wedge of three circles, and the tripod is contractible. $\square$

Remark 7.9 (Anatomy of the flat leak). (a) *The corner mechanism.* For an acute triangle the failure is isolated exactly where Remark 7.7(b) did not look. Each single-cell attachment is trivial: $W \cap H_e$ is the strip between the two corner cones flanking e , contractible, as is H_e . But two cells adjacent at a vertex p overlap in the nonempty open cone $H' \cap H'' = p + \{v : \langle v, n' \rangle > 0, \langle v, n'' \rangle > 0\} \subseteq p + N_p \subseteq \mathcal{A}^*(X)$ (the containment because the normals subtend an angle $\geq 90^\circ$): the double overlap sinks *entirely* into the Aubry set, $W \cap H' \cap H'' = \emptyset$, and the finite-intersection attachment fails as badly as possible. The cover $\{W, H_1, H_2, H_3\}$ is good (all nonempty pieces convex or strips; $H_1 \cap H_2 \cap H_3 = \emptyset$ since the weighted outward normals sum to zero), so the nerve theorem applies: the nerve is the complete graph K_4 — four contractible vertices, six edges, every triangle missing — which is a wedge of three circles. Theorem 7.5 thus computes $\mathcal{R}_X$ exactly; it is $W \hookrightarrow \mathcal{R}_X$ that fails. The three leaked classes are localised by winding: a loop winding once about p must cross the unbounded cone $p + N_p \subseteq \mathcal{A}^*(X)$, so no captured loop winds about any vertex, while the cycle $W \rightarrow H' \rightarrow H'' \rightarrow W$ around p winds once. For an *obtuse* triangle the cover is not even good: the obtuse corner's overlap cone protrudes past its normal cone, the protrusion is captured near the apex and Aubry far out (inside the opposite vertex's cone), and both $W \cap H' \cap H''$ and one $W \cap H_e$ become disconnected — the goodness threshold of item (5b) of §8 is visible already for triangles: the ideal cover of three points is good precisely when no angle is obtuse.

(b) *The leak count for finite sets.* Let $X \subset \mathbb{R}^2$ be finite and affinely spanning, $k = |X|$, with no point of X in the relative interior of a hull edge, and let V be the number of hull vertices. Consecutive hull vertices bound an unbounded Voronoi edge, so every hull edge is defective as above; the corner computation of Proposition 7.8 covers every normal cone, and $\mathcal{R}_X = \mathbb{R}^2 \setminus X \simeq \bigvee_k S^1$. On the other side M_X is the Voronoi skeleton, whose first Betti number is the number of *bounded* Voronoi cells, i.e. of interior points of X . The leak therefore has rank exactly $V \geq 3$: one class per hull corner. Complete reconstruction is generic for finite planar sets; what fails, by exactly the number of Aubry corners, is transport.

(c) *Corners are the mechanism.* The notched half-plane $X = \{y \leq 0, |x| \geq 1\}$ has $\mathcal{A}^*(X) = \{y > 0, |x| \geq 1\}$, the vertical rays over the shelf, and no hull corner; the single ideal cell $\{y > 0\}$ (centres $(0, t)$, feet $(\pm 1, 0)$) covers the Aubry set at level 0, the reconstruction is complete, and there is no leak: $\mathbb{R}^2 \setminus X$ and $M_X = \{x = 0\}$ are both contractible. Separately, the *single-cell* attachment can itself fail further afield: for $X = \text{unit circle} \cup \text{the plane } \{z = -10\}$ in $\mathbb{R}^3$, the cell $H = \{z > 0\}$ is defective (centres on the circle–plane bisector bowl escape with $\psi \rightarrow -z$), its Aubry trace is the cylinder shell $\{\rho = 1, z > 0\}$, and $H \cap W = (\text{open tube}) \sqcup (\text{outside})$ is disconnected — though there the reconstruction is incomplete below the plane.

(d) *The moral, inverted.* The leak of Remark 6.6 is not a curvature phenomenon: it is affine, generic, and quantified by the corners of the convex hull. What the cylinder adds is only that the cells themselves can be homotopy circles; the flat leak builds its loops from contractible cells glued around a corner. Lieutier's theorem is genuinely a theorem about bounded domains — completeness of the reconstruction does not restore it in the unbounded setting — but the medial complex does, everywhere (Theorem 7.5).

8 What remains

The theory is now unconditional in its dynamical form on every complete Riemannian manifold, and, with §7, in its homotopy form as well. What remains open is exactly the *faithful* layer — certificates decidable from X and the geometry of M , with M_X^g absent — together with the regularity of the medial complex; all of it lives on the non-compact (escape) side.

- (1) *Certificates in mixed curvature.* For which ideal points (horofunctions) h of a general non-compact M is $h + c$ a medial limit function, read from the contact structure of X on $\{h = 0\}$? Known: $\text{conv}/\text{Motzkin}$ in $\mathbb{R}^n$ [1, 28]; hconv on Hadamard manifolds, with the Carnot and Alexandrov frontiers of [4, §9–10]; vacuous on compact M ; and now the flat cylinder, whose end certificate is the sharp dichotomy of Proposition 5.1. Unknown already for a complete surface with both signs of curvature; and unwritten for the cusped hyperbolic surface (horocyclic cross-sections shrinking up the cusp) and the general flat quotient $\mathbb{R}^n/\Gamma$ (toral and mixed ends), which should follow the cylinder’s two steps — horofunction ends, then a wrap-around forcing argument — but are outside the theorems as stated.
- (2) *The level function.* On an inhomogeneous Hadamard manifold, characterise $c(\xi) = \inf\{c : b_\xi + c \text{ medial limit}\}$ from X alone (Remark 4.6); $c(\xi) = 0$ under pinching, in the flat case, and over any bisector witness, and no example with $c(\xi) > 0$ is known — deciding whether the deep-part phenomenon of [4, Thm. 3.4] actually occurs is the sharpest open question on the Hadamard side.
- (3) *The two-point axis in variable curvature.* Quantify cut-shielding: the prolate threshold $c^* = 2.6 \pm 0.1$ awaits an analytic determination, and more generally the class of compact M with $\mathcal{R}_X = M \setminus X$ for all sparse X (constant curvature is sufficient; bounded eccentricity conjecturally) [5, 6].
- (4) *Chebyshev problems.* The detection questions the series isolated on its frontiers: Motzkin in Carnot groups ($\mathbb{CH}^n$ horospheres) and in Alexandrov spaces of indefinite curvature (pinched horospheres) [4, §10]; and the domain (Lieutier) problem on $\mathbb{S}^n$, where the boundary’s positively curved intrinsic geometry becomes the detection object [6, §7].
- (5) *Regularity of the medial complex.* The homotopy question across the Aubry set, open in an earlier draft of this note, is settled by §7: the medial complex — M_X^g with the ideal free horoballs glued on — always carries the homotopy type of $\mathcal{R}_X^g$ (Theorem 7.5), because the finite layer does not leak (Theorem 7.3). What remains is its structure. (a) *The flat leak, localised:* the no-leak conjecture of the earlier draft is settled, negatively — three points in the plane reconstruct completely ($\mathcal{R}_X = \mathbb{R}^2 \setminus X$, the sets characterised by [1, Thm. 6]) and leak all three of their loops past a contractible axis (Proposition 7.8); for finite planar X the leak has rank exactly the number of hull corners, and it dies without corners (Remark 7.9). What survives is localisation: is the leak in $\mathbb{R}^n$ likewise carried by the non-smooth strata of $\partial \text{conv } X$, and which pairs of homotopy types $(\mathbb{R}^n \setminus X, M_X)$ are realised by completely reconstructing sets? (b) *A CW or nerve model:* when is the ideal cover good, so that the nerve theorem [13] computes the complex (for three points: iff the triangle has no obtuse angle, Remark 7.9(a)) — and can the local-tree and delta-convex structure theory of [31, 27], or Damon’s skeletal calculus [18], produce a finite model in the generic compact-contact case? (c) *Grazing rays:* can $\mathcal{A}^*(X) \cap \text{Cut}(X) \neq \emptyset$ actually occur

for a closed (necessarily non-smooth, Proposition 6.8) contact set? Lemma 7.2 shows the theory is indifferent, but the example, or its impossibility, would calibrate how wild the transition between capture and escape can be.

Conclusion. Białożyty’s Theorem 6 generalises to every complete Riemannian manifold, in two layers. The *unconditional* layer is finished here: reconstruction by the two-geodesic axis is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3); every point is captured, swallowed, or escaping (Theorem 4.4); capture is total on compact manifolds (Corollary 3.2), with no hypotheses whatsoever beyond completeness and closedness. The *faithful* layer — replacing “is a medial limit” by a certificate on contact sets — is exact on the flat knife-edge (conv, [1]), on every Hadamard manifold (hconv, [4]), on the flat cylinder (the end dichotomy of Proposition 5.1), and trivial on compact manifolds; its extension into mixed curvature, together with the Chebyshev-set problems it generates, is the genuine residue. The gaps flagged in the series — the containment step past cut points and the radial offset — are closed, the first by stopping the ascent at its first approach to the axis instead of flowing through it, the second by normalising the limit objects by their own reach. A topological layer now sits on top: the captured part of the complement always has the homotopy type of the axis (Theorem 6.5, via the Aubry dictionary of Lemma 6.4), totally so when $\sup d_X < \infty$ (Corollary 6.3); the flat cylinder separates the two completions — swallowing finishes the reconstruction across the Aubry set, but only capture transports homotopy type — and the medial complex re-fuses them: the finite layer does not leak (Theorem 7.3), so the axis with its ideal free horoballs glued on carries the homotopy type of $\mathcal{R}_X^g$ on every complete manifold, for every closed contact set (Theorem 7.5); the leak the complex repairs is already flat — three points in the plane leak all of their homotopy past a contractible axis, one loop per hull corner (Proposition 7.8) — and the residue of the homotopy programme is regularity, not existence (§8(5)). In the smooth case the layer is classical geometry made medial: the captured part is the mapping cylinder of the swallow map onto the cut locus, its collapse is the Thom space [10], and its uniform-capture instances are the Ford–Voronoi and crystallographic spines (Corollary 6.7, Proposition 6.8).

References

- [1] A. Białożyty, *Sets reconstructable with medial axis*, arXiv:2604.20671 (2026). (§5.2 poses the compact Riemannian reconstruction question settled by Corollary 3.2.)
- [2] *Sets reconstructable with medial axis: a generalisation to Riemannian manifolds*, companion note, June 2026 (file `medial_axis_riemannian_1.pdf`).
- [3] *Medial-axis reconstruction in hyperbolic space, with a structural framework for Hadamard manifolds*, companion note, June 2026 (file `medial_axis_riemannian_2.pdf`).
- [4] *Medial-axis reconstruction in Hadamard manifolds: a faithful, curvature-free characterization*, companion note, June 2026 (file `medial_axis_riemannian_3.pdf`).
- [5] *Medial-axis reconstruction in positive curvature: the focal collapse and total reconstruction*, companion note, June 2026 (file `medial_axis_spherical.pdf`).

- [6] *Two axes of medial-axis reconstruction: compactness, and the Busemann–Hessian detection trichotomy*, companion note, June 2026 (file `medial_axis_trichotomy.pdf`).
- [7] J.-P. Aubin and A. Cellina, *Differential Inclusions*, Grundlehren 264, Springer, 1984. (Existence and extension for upper semicontinuous convex compact right-hand sides, §2.1.)
- [8] P. Albano and P. Cannarsa, *Structural properties of singularities of semiconcave functions*, Ann. Scuola Norm. Sup. Pisa **28** (1999), 719–740.
- [9] P. Albano, P. Cannarsa, K. T. Nguyen and C. Sinestrari, *Singular gradient flow of the distance function and homotopy equivalence*, Math. Ann. **356** (2013), 23–43.
- [10] S. Basu and S. Prasad, *A connection between cut locus, Thom space and Morse–Bott functions*, Algebr. Geom. Topol. **23** (2023), 4185–4233.
- [11] A. Bhowmick, *Stability of the injectivity radius and the cut locus of submanifolds under perturbations*, arXiv:2605.26499 (2026).
- [12] R. L. Bishop, *Decompositions of cut loci*, Proc. Amer. Math. Soc. **65** (1977), 133–136.
- [13] K. Borsuk, *On the imbedding of systems of compacta in simplicial complexes*, Fund. Math. **35** (1948), 217–234.
- [14] M. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Grundlehren 319, Springer, 1999. (Horofunctions and Busemann points, II.8.)
- [15] P. Cannarsa, W. Cheng and A. Fathi, *Singularities of solutions of time dependent Hamilton–Jacobi equations. Applications to Riemannian geometry*, Publ. Math. Inst. Hautes Études Sci. **133** (2021), 327–366.
- [16] F. Chazal and A. Lieutier, *The λ -medial axis*, Graphical Models **67** (2005), 304–331.
- [17] J. Cheeger, *Critical points of distance functions and applications to geometry*, in: Geometric Topology: Recent Developments, Lecture Notes in Math. 1504, Springer, 1991, 1–38.
- [18] J. Damon, *The global medial structure of regions in $\mathbb{R}^3$* , Geom. Topol. **10** (2006), 2385–2429.
- [19] D. Dugger and D. C. Isaksen, *Topological hypercovers and $\mathbb{A}^1$ -realizations*, Math. Z. **246** (2004), 667–689.
- [20] D. B. A. Epstein and R. C. Penner, *Euclidean decompositions of noncompact hyperbolic manifolds*, J. Differential Geom. **27** (1988), 67–80.
- [21] V. Gershkovich and H. Rubinstein, *Morse theory for Min-type functions*, Asian J. Math. **1** (1997), 696–715.
- [22] M. Gromov, *Hyperbolic manifolds, groups and actions*, in: Riemann Surfaces and Related Topics, Ann. of Math. Stud. 97, Princeton, 1981, 183–213. (The horofunction compactification.)

- [23] K. Grove, *Critical point theory for distance functions*, in: Differential Geometry: Riemannian Geometry, Proc. Sympos. Pure Math. **54**, Part 3, Amer. Math. Soc., 1993, 357–385.
- [24] K. Grove and K. Shiohama, *A generalized sphere theorem*, Ann. of Math. **106** (1977), 201–211.
- [25] A. Lieutier, *Any open bounded subset of $\mathbb{R}^n$ has the same homotopy type as its medial axis*, Computer-Aided Design **36** (2004), 1029–1046.
- [26] C. Mantegazza and A. C. Mennucci, *Hamilton–Jacobi equations and distance functions on Riemannian manifolds*, Appl. Math. Optim. **47** (2003), 1–25. (Semiconcavity and first variation of the distance from a closed set.)
- [27] T. Miura and M. Tanaka, *Delta-convex structure of the singular set of distance functions*, Comm. Pure Appl. Math. **77** (2024); arXiv:2204.10449.
- [28] T. Motzkin, *Sur quelques propriétés caractéristiques des ensembles bornés non convexes*, Atti Accad. Naz. Lincei **21** (1935), 773–779.
- [29] S. Prasad, *Cut locus of submanifolds: a geometric and topological viewpoint*, arXiv:2303.14931 (2023).
- [30] T. Sakai, *Riemannian Geometry*, Transl. Math. Monogr. 149, Amer. Math. Soc., 1996.
- [31] M. Tanaka and S. V. Sabau, *The cut locus and distance function from a closed subset of a Finsler manifold*, Houston J. Math. **42** (2016), 1157–1197.